# West Bengal State Electricity Distribution Company Limited

**(*A Govt. Of West Bengal Enterprise*)**

## TECHNICALSPECIFICATION FOR 12kV, 800A, 18.4kA, MULTI PANEL (Incoming, Bus Coupler with Adaptor Paneland Outgoing) SHUNT TRIP, INDOOR TYPE SCADA COMPATIBLE SWITCHGEAR WITH MOTOR OPERATED SPRING CLOSING VACUUM CIRCUIT BREAKER

**April'2024**

## 1.0 CODES&STANDARDS:

Ratings, characteristics, tests and test procedures, etc. for the 11kV metal-clad switchgear panels and all the protection relays, measuring and indicating instruments and the control and monitoring devices and accessories, including current transformers and voltage transformers shall comply with the provisions and requirements of the standards of the International Electrotechnical Commission (IEC), IS where specified.

The latest revision or edition in effect at the time of bid invitation shall apply. Where references are given to numbers in the old numbering scheme from IEC it shall be taken to be the equivalent number in the new five-digit number scheme. The bidder shall specifically state the precise standard, complete with identification number, to which the various equipment and materials are manufactured and tested. The bid document may not contain a full list of standards to be used, as they only are referred to where useful for clarification of the text.

| Standard Name / No | Standard's Description |
|---|---|
| Indian Electricity Rules 1956 | Latest edition |
| Indian electricity act 2003 | Latest edition |
| Switchgear and control gear | IEC: 60694, IEC: 60298, IEC: 62271-200, IEC: 60529. IS: 3427, IS 12729, IS 12063, IS:13947, IS: 9046 |
| Circuit Breaker | IEC 62271-100, IS 13118, IS 2516 |
| Vacuum Interrupter | IEC 60056 |
| Current Transformers | IEC:60185, IS 16227 (PART 1 & 2) |
| Voltage Transformer | IEC:60186, IS 16227 (PART 1 & 3) |
| Indicating Instruments | IS:1248 |
| Energy Meters | As mentioned in Technical Specification of Energy Meter |
| Relays | IS 8686, IS 3231, IS 3842 |
| Control switches and push buttons | IS 6875 |
| Electromagnetic Compatibility | IEC 61000 |
| Arrangement of switchgear bus bars, main connection and auxiliary wiring | IS 375 |
| Code of practice for phosphating iron & steel | IS 6005 |
| Colours for ready mixed paints | IS 5 |

## 2.0 Scope of Work:

This Specification covers the design and engineering, manufacture, testing at the manufacturer's factory, painting, packing for transport, insuring, transportation by road and delivery at destination point of 12 kV, 800 A, 18.4kA, Multi panel, shunt trip, Indoor Type, motor operated, spring closing Vacuum Circuit Breaker having SCADA compatible facilities and complete with all accessories as specified hereinafter.

### 2.1 UNIT OF MEASUREMENT AND LANGUAGE

In all correspondence, in all technical schedules and on all drawings prepared by the manufacturer, the metric units of measurement shall be used. On drawings or printed pamphlets where other units have been used, the equivalent metric measurements shall be added. All documents, correspondence, drawings, reports, operating and maintenance instructions/manuals and nameplate details of the equipment shall be in English language.

## 3.0 SERVICE CONDITIONS:-

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| Max. Ambient Temperature | 50°C |
| Max. Ambient Temperature for design purpose | 40°C |
| Min. Ambient Temperature | 4°C |
| Max. Relative Humidity | 100% |
| Average no. of rainy days per annum | 100 days |
| Elevation | within 1000M from MSL |
| Maximum Wind Pressure | 150kg/m2 |
| Horizontal Acceleration due to seismic force | 59/100 |
| Pollution Level | Heavily polluted atmosphere |

From the geographical condition, the area where the switchgear panels shall be installed is categorized into the tropical climate zone. In choosing materials and their finishes, due regard shall be given to the humid tropical conditions under which the switchgear panels shall be called upon to work. The manufacturer of the switchgear panels shall submit details of his usual practice of tropicalization which have proven satisfactory for application to the switchgear panels and associated equipments to prevent rusting and ageing in the tropical climate zone. The applicable standards for tropicalization shall be listed.

**Climatic condition:** The equipment is required to operate satisfactorily under the following site conditions:

The switchgear panels shall be installed in a room without air conditioning but with ventilation to allow natural cooling. Therefore, all the protection and control devices employed shall be capable of operating in this environment without failure for their designed life time. Particularly the power supply modules of the protection and control devices shall be designed for minimum heat generation and effective heat dissipation to ensure that the temperature of these devices enclosed in the relay panels at the above listed ambient temperatures shall not exceed the maximum operating temperature of the device.

### 3.1 Tropicalization

(a) All equipment must be designed for operations in the severe tropic climate conditions and fully comply with climatic aging tests as per IEC 60932-class 2.

In choosing materials their finishes, due regard shall be given to the humid tropical conditions under which the switchgear will be called upon to work. The manufacturer shall submit details of his usual practice which have proven satisfactory for application to the parts of the Switchgear panels, which may be affected by tropical conditions.

**(i) Metals:**

Iron and Steel are generally to be painted or galvanized as appropriate. Indoor parts may alternatively have chromium or copper-nickel plates or other approved protective finish.
Small iron and steel parts (other than rustless steel) of all instruments and electrical equipment, the cores of electromagnets and the metal parts of relays and mechanisms shall be treated in an appropriate manner to prevent rusting.

**(ii) Screws, Nuts, Springs, e.t.c.:**

The use of iron and steels shall be avoided in instruments and electrical relays wherever possible. Steel screws shall be zinc, cadmium or chromium plated or where plating is not possible owing to tolerance limitations, shall be of corrosion resisting steel. Instrument screws (except those forming part of a magnetic circuit) shall be of brass or bronze. Springs shall be of non-rusting material, e.g., phosphor-bronze or nickel silver, as far as possible.

(iii) **Rubbers:**

Neoprene and similar synthetic compounds, not subject to deterioration due to the climatic conditions, shall be used for gaskets, sealing rings, diaphragms, etc.

### 3.2 WORKING STRESS AND EQUIPMENT / APPARATUS DESIGN

#### 3.2.1 General

a) The design, dimensions and materials of all parts shall be such that they will not suffer damage under the most adverse conditions nor result in deflections and vibrations, which might adversely affect the operation of the equipment. Mechanisms shall be constructed to avoid sticking due to rust or corrosion.

b) The equipment and apparatus shall be designed and manufactured in the best and most substantial and workmanlike manner with materials best suited to their respective purpose and generally in accordance with up-to-date recognized standards of good



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c) Whenever possible, all similar parts, including spare parts, shall be made interchangeable. Such parts shall be of the same materials and workmanship and shall be constructed to such tolerances as to enable substitution or replacement by spare parts easily and quickly.

d) All equipment shall be designed to minimise the risk of fire and consequential damage, to prevent ingress of vermin and dust and accidental contact with electrically energized or moving parts. The switchgear panels shall be capable of continuous operation with minimum attention and maintenance in the exceptionally severe conditions likely to be obtained in a tropical climate and where the switchgear is called upon to frequently interrupt fault currents on the system and also where the duty of operation is high.

**3.2.2 Strength and quality**

a ) All steel castings and welding shall be stress-relieved by heat treatment before machining, and castings shall be stress-relieved again after repair by welding.

b) Liberal factors of safety shall be used throughout, especially in the design of all parts subject to alternating stresses or shocks.

**3.2.3 Designed data for low voltage equipment**

Low voltage equipment and installation shall be designed in accordance with EMC(Electromagnetic Compatibility, IEC 61000) directives. The rating and design criteria for low voltage equipment shall be as follows:

**AC Supply Rating system**

i. Rated voltage between phase 415 V AC
ii. Connection type 3ph 4wire
iii. Rated voltage between phase to earth 240 V AC
iv. Grounding system PME
v. Frequency 50 HZ
vi. Voltage variation ± 10 %
vii. Frequency variation ± 5 %
viii. Power frequency 1 min, Test Voltage 3 kV
ix. Thermal rating of conductors 120 % of load

The AC supply shall be used for power circuit and for lighting, indication, motor controls and similar small power circuits. Unless otherwise specified, the equipment provided under this tender is to be capable of reliable operation at voltages as low as 85% of the rated voltage, and to withstand continuously up to 110% supply voltage above the rated value of 240V or 415VAC.

**DC Auxiliary Supply Rating**

i. Equipment/Device Rated voltage 30V DC
ii. Connection type 2 wire.
iii. Voltage variation 24 to 40 V DC

The auxiliary DC supply shall be used for controls, indication, alarm, protection relays, and circuit breaker tripping and closing circuit, etc. All equipment and apparatus including the circuit breakers, protective relays, control devices and accessories, measuring and indicating instruments and electronic equipment shall be capable of satisfactory operation at 80% to 130% of the rated DC supply voltage. However, in case of VCB, for tripping the range should be 70% to 110% and for closing that should be 85% to 110%.

### 3.2.4 Electrical controls, auxiliaries and power supplies

**(a) Responsibility for electrical control and auxiliaries.**

The manufacturer shall provide all control, indication, alarm and protection devices and all auxiliary equipment with wiring and interconnecting cable which are integral parts of or are directly associated with or mounted on the switchgear panels to be supplied under this tender. The design of protection and control schemes for the switchgear panels shall be subject to approval of WBSEDCL.

b) **Operation and control.**

Interlocking devices shall be incorporated in the control circuit to ensure safety, and proper sequence and correct operation of the equipment. The scheme will be finalised during detailed engineering and drawing approval.

### 3.2.5 Corona and radio interference

a) Switchgear shall electrically be designed to avoid local corona formation and discharge likely to cause radio interference.

b) The design of jointing of adjacent metal parts and surfaces shall be such as to prevent corrosion of the contact surfaces and to maintain good electrical contact under service conditions.

c) Particular care shall be taken during manufacture of busbar and fittings and during subsequent handling to ensure smooth surface free from abrasion. All joints on the busbar and the circuit within the switchgear board shall be silver or tin-plated to ensure good electrical connection.

## 4.0 PANELCONSTRUCTION

Intheeventofdirectconflictbetweenvariousorderdocuments,theprecedenceofauthorit yof documents shall be as follows –

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| 4.1 | **Enclosure Type** | Dead front, floor-standing, rigid welded steel frames fully compartmentalized, Metal clad, Vermin Proof, suitable for indoor installation and provision for bolting to the floor. |
| 4.2 | **Enclosure degree of protection** | IP 5Xfor High Voltage compartment and IP4X for Low Voltage compartment |
| 4.3 | **Enclosure Material** | CRCA steel |
| 4.4 | **Load bearing members** | Minimum 2.5mm thick |
| 4.5 | **Doors and covers** | **Minimum 2.0mm thick. Floor to circuit breaker door clearance minimum 15 mm shall be maintained so that Insulation mat can be placed.** |
| 4.6 | **Gland Plate (detachable type)** | 3.0mm MS detachable type for 3 core cable and aluminium 5.0mm for single core cables. Cable compartment shall have an anti vermin guard plate for protection against entry by rats, rodents etc. |
| 4.7 | **Maximum Dimension of the Panel (WxDxH)** | **700mmx2000mmx2700mm.** |
| | **Maximum operating height of the panel** | **Operating height 1800 mm (max).** |
| 4.8 | **Extensibility** | The Switchgear shall be designed so that future units can be added to each sides (unless coupled to other equipment). A removable plate will cover any unused openings in the side of the panel. |
| 4.9 | **Extension Bus and Bus Wires** | **In case of Single feeder panels, Identical Bus Bars and Bus Wires for connection with adjacent Incomer / Bus Coupler / Feeder needs to be supplied.** |
| 4.10 | **Separate compartment for** | 1. Bus bar,2. circuit breaker, 3. HV incoming / outgoing cable, 4. LV instruments relays,**5. Metering and SCADA terminal at top of the panel** |



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| 4.11 | **Breaker compartment door** | **Breaker chamber door should be hinged type with suitable locking arrangement. Breaker chamber door with bolted design is not acceptable**(design with breaker trolley as the front cover is not acceptable) |
| 4.12 | **Breaker to bus bar compartment** | Through seal off bushings/Spout |
| 4.13 | **Breaker to cable** | Through seal off bushings/Spout |
| 4.14 | **Pressure relief devices** | To be provided for each HV Compartment. Each compartment shall be separated from adjacent one by sheet steel barrier. |
| 4.15 | **Bus support insulator** | Non hygroscopic, track-resistant, high strength, epoxy insulators (calculation for validating dynamic force withstand capability to be submitted during detailed engineering) |
| 4.16 | **Fixing arrangement**<br>**i. Doors**<br>**ii. Covers**<br>**iii.Gasket** | Concealed hinged Bolted with SS bolts and Neoprene rubber gasket , washer |
| 4.17 | **Required HV cable termination height in the cable compartment and cable termination barrier** | 550mm for 11 kV<br><br>**Suitable 4 nos FRP phase Barriers made of non hygroscopic material preferably Fibre Glass sheet should be provided in cable chamber between phases and phase to earth.** |
| 4.18 | **Panel Base Frame** | Steel base frame as per manufacturer's standard. |
| 4.19 | **Handle** | Removable bolted covers for cable chamber and busbar chamber shall be provided with "C" type handles |
| 4.20 | **Prevention of Internal Arc** | Shall be type tested against internal arc as per provision in IEC 62271-200. The Circuit Breaker, busbars and cable compartments shall be provided with arc venting outlet. The doors for the compartment shall be capable of withstanding the effects of maximum internal arcing fault without being blown off and causing danger to personnel and other equipment. This should be proven by successful testing for 18.4 KA with duration 0.1 second as per relevant IEC standard. **Higher rated panel may also be accepted. In all cases manufacturer has to offer their exact type tested design panels. Breaker chamber door should be hinged type with suitable locking arrangement. Breaker chamber door with bolted design is not acceptable.** |

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| 4.21 | **Test Terminal Block** | 13 Ways, 50A, 1100V with back connection. Dia of Entries is 5 mm & overall dimension 250X50 mm, covered with insulated cover. TTB should be Disconnecting Link type suitable for 3 Phase 4 Wire Energy Meter. Screw type TTB is not acceptable. |
| 4.22 | **Multiway terminal Block and low voltage wiring** | Delinking type, Rail/Channel mounted, Terminal Connector to be used in CT Circuit & Screw type for other Circuit. The Terminal Blocks should be suitable for 2.5 sq.mm wire and covered with insulated transparent cover. Pitch should be minimum 8mm & 10mm for Screw type & Delinking type connectors respectively. The low voltage cable shall be enclosed in grounded metal conduit when routed through a high voltage compartment. Control wiring shall be neatly bundled and tie wrapped where applicable. Wiring shall be protected from rubbing against door flanges or other parts of the enclosure. |
| 4.23 | **Cable Tray** | Netted Metal cable Tray of suitable size at the rear side of Switchgear, preferably running at the top along the panel for carrying the signal cables for SCADA interface to be provided. |
| 4.24 | **Remote Terminal Box** | To be provided suitably at the back side of the panel for termination of the following cables :<br>A . Main AC / DC<br>B . Inter trip from up stream<br>C . SCADA related terminations<br>D . Inter connection for Differential protection etc. |
| **4.24.1** | **Voltage Presence Indicator (VPI)** | **a. Live line Indicators: - Capacitive voltage indicators shall be provided on feeder side in outgoing and incoming feeders to indicate the voltage presence in each phase. Voltage Presence Indicator (VPI) to be placed at suitable rear/front side of the panels (Position to be finalised during drawing approval stage).**<br>**b. The panel shall have Voltage Presence Indicator (VPI) to warn the operator against earthing of live connections.**<br>**c. Use of voltage presence indicator sensors having high linearity over the entire working range and incorporating potential / capacitive divider circuit for connection to voltage indicator lamps.** |
| 4.25 | **Circuit Breaker** | |
| 4.25.1 | Mounting | On withdraw able truck/trolley or carriage, with locking facility in service position. Switchgear truck/trolley should be floor mounted. Racking-in and Racking-out should be such that one person |



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| 4.25.2 | Switching duty | a) Transformer oil filled and dry type<br>b) Under ground cable with upto 10KM. |
| 4.25.3 | Interrupting medium | Vacuum |
| 4.25.4 | Breaker operation | Three separate identical single pole units operated through the common shaft and shall be fully interchangeable both electrically and mechanically. Circuit breaker poles between the interrupters and primary plug-in contacts shall be fully insulated with durable material. Each breaker shall be provided with Mechanical 'ON' and 'OFF' facility by operating suitable closing and opening devices. each breaker shall be provided with Mechanical 'ON' and 'OFF' indicators. Each breaker shall have three positions- service, test and isolated/withdrawal marked. mechanical safety interlocks shall be provided so that it is not possible for a circuit breaker<br>a. To be put into the cubicle unless the truck is secured in position.<br>b. To be either draw out or draw in from and to the service position unless its contacts are safely open.<br>c. To be withdrawn or inserted in the fixed housing unless it is at the withdrawable position.<br>d. To be operated in service position unless its primary and secondary isolating contacts are fully engaged.<br>e. Rack In and Rack out of the VCB trolley possible when the Breaker door in close position. The interlock should be clearly mentioned on the front door of the Breaker.<br>f. The circuit breaker racking equipment can be padlocked in closed position.<br>g. Electrical close/trip operation should be dependent on Local/Remote switch. However, protection trip and |
| 4.25.5 | Operating mechanism | Re-strike free,tripfree both electrically and mechanically, with electrical anti-pumping feature |
| | | One O-C-O operation possible after failure of power supply to the spring charging motor. |
| | | Motor wound, spring, charged, stored energy type with Manual charging facility. One no. Breaker Truck operating handle for every three panels needs to supply. |
| 4.25.6 | Trip and Closing Coil | To be rated for substation DC voltage. Suitable for operation at minimum operating voltage of 70% for tripping and 85% for closing operation. Maximum Burden shall be about 250 watt for each coil. |

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| 4.25.7 | Truck | The circuit breaker shall be mounted on an inbuilt carriage to facilitate isolation and withdrawal of the breaker. **Trolley of the VCB should be horizontally isolated** and **there should be a minimum ground clearance of 0.55 meter between thimble point of cable termination and GL. There should be a minimum clearance of 300 mm between any 11kV live bus and GL.**<br>Trolley should be detachable from panel without any additional supporting trolley i.e. VCB trolley itself move on ground level. |
| 4.25.8 | Adaptor Panel | Separate Adaptor Panel with each Bus Coupler panel shall be designed. |
| **4.26** | **Breaker Indications and push buttons** | |
| 4.26.1 | ON/OFF/Emergency trip push button | i) Electrical and Mechanical type in front of the panel<br>ii) Mushroom type Emergency Off push button will be provided with a protective flap in front of the panel. |
| 4.26.2 | Mechanical ON-OFF indication | On breaker trolley front |
| 4.26.3 | Operation counter | On breaker trolley front |
| 4.26.4 | Test-service position indicator | On the front of the panel |
| 4.26.5 | Mechanism charge/discharge indicator | On breaker trolley front |
| 4.26.6 | Breaker positions | Service, testand isolated |
| 4.26.7 | **Inter changeability** | **The Circuit Breakers of Incomer, Bus Coupler & Feeder Panels should be interchangeable.** |
| 4.26.8 | Breaker control | On panel front only |
| 4.26.9 | Handle | Breaker shall be provided with handles for easy handling, rack in-out operation and manual spring charging as applicable. |
| 4.27 | **Functional Requirements** | |
| 4.27.0 | Interlock and safety devices | |
| 4.27.1 | Breaker compartment door opening | Cannot be opened unless breaker is OFF and racked out to test/ isolated position |
| 4.27.2 | Breaker compartment door closing | Should be possible even when breaker in isolated position |
| 4.27.3 | Racking mechanism safety interlock | Mechanical type |
| 4.27.4 | Racking in or out of breaker inhibited | When the breaker is closed |
| 4.27.5 | Racking in the circuit breaker inhibited | Unless the control plug is fully engaged |
| 4.27.6 | Disconnection of control plug inhibited | As long as the breaker is in service position |



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| 4.28 | **Additional Requirement** | |
| 4.28.1 | Exposure to live parts | In case the breaker panel door is required to be opened during a contingency, the personnel should not be exposed to any live parts. Suitable shrouds / barriers / insulating sleeves should be provided as required. |
| 4.28.2 | a. Operation of breaker in Local<br>b. Operation of breaker in Remote<br>c . Protection Trip | a. In either service or test position<br>b. Only in Service Position<br>c . Both in Local and Remote position of LR Switch. |
| 4.28.3 | Closing from local | Only when local/remote selector switch is in local position |
| 4.28.4 | Closing from remote | Only when local/remote selector switch isin remote position |
| 4.28.5 | Tripping from local | Selector Switch in Local |
| 4.28.6 | Tripping from remote | Selector switch in 'Remote' position |
| 4.28.7 | Testing of breaker | In test or isolated position keeping control plug |
| **4.29.** | **Safety shutters** | |
| 4.29.1 | Automatic safety shutter for primary disconnector | To fully cover contacts when breaker is withdrawn to test.<br>Independent operating mechanism for bus bar & cable side shutters, separately pad lockable in closed position. |
| 4.29.2 | Label for identification | For bus side and cable side shutters |
| 4.29.3 | Warning label on shutters of incoming and other connections | Clearly visible label "isolate elsewhere be for earthing" be Provided |
| **4.30** | **Breaker electrical operation features** | |
| 4.30.1 | Trip coil supervision relay | To be given for breaker close and open condition. |
| 4.30.2 | Trip coil supervision relay contact | For indication and alarm |
| 4.30.3 | Master Trip Relay (NO) Contact | Wired directly to trip coil |
| 4.30.4 | Master trip relay (NC)contact | Wired to inhibit closing of breaker |
| 4.30.5 | Emergency Trip Push button contact | Wired directly to trip coil of breaker |
| 4.30.6 | DC Control supply and AC supply bus in all panels | Fed by single DC and AC Incoming source in buscoupler panel as well as all Incomers in case of total Switch Board. Isolating fuse units shall be provided for incoming supplies to each switchgear unit. |
| 4.30.7 | PT supply bus in all panels | Fed normally by Bus PT with changeover facility. |



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| 4.30.8 | **Spring Charge Limit switch Aux. Contact Configuration** | The circuit breaker shall be provided with motor operated spring charged closing. Spring charging by motor should be smooth and hassle free and there should be nominal sound during spring charging. Motor should be 230 Volt AC operated, Tripping of the circuit breakers shall be through "Shunt trip" coils rated for 30V DC auxiliary supply. It shall be possible to trip the breaker manually in case of necessity.<br><br>2NC --- For Motor circuit<br>1 NO --- For Breaker Closing Circuit<br>1 NO --- For Breaker Closing permissive to be used in IED/Relay.<br>1 NO --- For panel spring charged indication<br>AC operated contactor may be used for contact multiplication of Limit Switch, if required, for indication and spare contacts. |
| 4.30.9 | **Local/ Remote selector switch** | Switch should be 4 pole in each position, 2 way lockable and stay put type. |
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| **5.1** | **Current Transformers** | Shall be cast resin type with insulation class of E or better. Contact tips on primary side shall be silver plated. Correct polarity shall be invariably marked on each primary and secondary terminal.<br>Primary shall be wound or bar type, rigid, high conductivity grade copper conductor. Unavoidable joints on the primary conductor shall be welded type, preferably lap type. Current density at any point shall not exceed 1.6 A/sq.mm. Suitable insulated copper wire of electrolytic grade shall be used for CT secondary winding. Multi ratio in CT shall be achieved by reconnection of secondary winding tapping. The secondary terminals shall have screw type terminals. The screw should have sufficient length for connection of at least two nos wires with plain and spring washers and minimum 10 mm clearance between the adjacent screw. |
| **5.2** | **CT Mounting** | **Fixing in horizontal upright condition over a base channel is preferable.**<br><br>**In case of CT base plate fixed at the upper side of the CT Chamber and fixing of CT in hanging condition, proper sliding channel for CT base plate holding shall be provided to facilitate easy replacement of 11 kV CT afterwards.**<br><br>**CT Secondary connection terminals should be clearly visible and accessible from the back side of the breaker assembly just after opening the rear cover plate.** |
| **6.0.0** | **Potential Transformer** | |



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| 6.1.0 | Type | Potential Transformer should be Three Phase Five Limb type or it should be combination of three number Single Phase PT housed in a withdraw able carriage. Shall be cast resin type with insulation class of E or better. Service position locking mechanism shall be provided and indicated by bidder in relevant drawing. Rigidity of primary star point with earth bus in service position shall be confirmed. Connection of Primary neutral with main earth bus to be made with solid copper of minimum 50 sq mm.<br>Contact tips of primary/secondary contacts shall be silver plated. Correct polarity shall be distinctly marked on primary and secondary terminal. Secondary terminal studs shall be provided with at least three nuts, two plain and two spring washers for fixing leads. The stud, nuts and washers shall be of brass, duly nickel plated. The minimum outside diameter of the studs shall be 6 mm. the length of at least 15 mm shall be available on the studs for inserting the leads. The space clearance between nuts on adjacent studs when fitted shall be at least 10 mm. |
| 6.2.0 | **Mounting** | **Rail mounted on top of the Unit and connected on bus side. It can be plugged into and withdraw from service by pulling or pushing the PT by the handle provided on the PT. This action traverses the PT along the rails and shall automatically operate the spout shutters. The shutter drive also forms a latch which holds the PT in the service position and this latch shall be required to be released before PT can be isolated.**<br>**Access to the PT and the reinforcement in the Panel for allowing a person to stand on the Top should be provided.**<br>**Locking arrangement of the PT with the carriage and secondary fuses are to be provided.**<br>**Mounting of PT on the breaker truck is not acceptable.** |
| **7.0.0** | **Equipment Earthing** | |
| 7.1 | Material of Earthing bus | Two separate earthing terminals shall be provided in each panel and shall be connected to the earth bus within the panel. The earth bus shall be of **tin plated** copper and shall have adequate cross sectional area. Earthing conductors shall be of annealed high conductivity stranded Copper in accordance with Table-4 in BS.6346 and protected with an extruded PVC sheath of 1100 volt grade. The earthing conductor on the primary equipment as well as for external connection to substation earthing grid shall be adequate to carry the rated switchgear short-circuit current of 18.4kA for 1 second. |
| 7.2. | Earth bus joints | A ground bus rated to carry maximum fault current shall be furnished along the full length of the panel board. Each stationary unit shall be grounded directly to ground bus. Allbolted joints in the buswillbe effected by connection oftwo bolts. |



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| 7.3. | Size | Rectangular Earth Bus of Minimum size 25mmX6mm |
| 7.4. | Enclosure and non-current carrying part of the switchboard/components | Effectively bonded to the earth bus |
| 7.5. | Hinged doors | Earthed through flexible copper braid |
| 7.6. | Circuit breaker frame / carriage | Earthed before the main circuit breakercontacts/control circuit contacts are pluggedin the associated stationary contacts. |
| 7.7. | Metallic cases of relays, instruments and other LT panel mounted equipment | Connected to the earth bus by independent copper wires of<br>size not less than 2.5 sq mm with green colour insulation. For this purpose LT compartment should have a clear designated earth bus to which earth connections from all componentsare to be connected. |
| 7.8. | CT & PT Secondary neutral | Earthed at one place at the terminal blocks through |
| **8.0** | **Indicating Meters** | Flush Mounted, back connected, dust proof with Industrial grade "A" classification and conforms to IS:1248(1968) |
| 8.1 | CT operated Ammeter | 3 no. Digital Ammeters to be provided in Incomer and Feeder panels for measuring load current of each phases.<br>Display : 3 ½ -digit display,<br>Aux. Power: : 230 V AC<br>CTR: 600/1A for Incomer and 200/1A for Feeders.(Site Selectable)<br>Accuracy Class: 1.0 or less.<br>Size : 96 X 48 mm. |
| 8.2. | Direct Ammeter | 1 nos. Separate Digital Ammeter to be provided for measuring heater current drawal in all the Panels.<br>Display : 3 ½ -digit display,<br>Aux. Power: : 230 V AC<br>Accuracy Class : 1.0 or less.<br>Size : 96 X 48 mm. |
| 8.3. | P T operated Voltmeter | One no. Digital Voltmeter to be provided in Incomer Panels for measurement of Phase to Phase system voltage, along with a Voltage Selector Switch.<br>Display : 3 ½ -digit display,<br>Aux. Power: : 230 V AC<br>PTR : 11000 / 110 V<br>Accuracy Class : 1.0 or less.<br>Size : 96 X 48 mm. |
| 8.4. | Direct Voltmeter | One no Analog DC Voltmeter to be provided in the Bus Coupler panel for measurement of Control DC voltage. |



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| **9.0** | **Energy meter** | One Three Phase, Four Wire CT/PT operated, fully static AMR compatible TRIVECTOR Energy Meter *along* with TTB to be provided in all Incoming and outgoing breaker panels at its upper most position in a separate metering chamber. Necessary wiring to be provided with separate terminal block (suitable for CT/PT connection) for Energy meter connection and as per detailed technical specification attached in a separate sheet (Annexure-F). TTB to be installed in the metering chamber. |
| **9.1** | **CB Contact multiplier** | One **30 V DC** operated contact multiplier (6 NC) to be provided for CB contact multiplication. Energy Meter PT supply to be routed through the contact of the contact multiplier. |
| **10.0** | **Multi functional Transducer** | Will be supplied by the Utility and to be fixed in the metering chamber. Separate TB for connection should be provided. Space required for Transducer fixing is 100X100mm. |
| **11.0** | **Indication** | Flush mounted, insulated from 30V DC supply with appropriate coloured lens. The lens shall be made of a material, which will not be softened by the heat from the lamps. |
| 11.1 | Lamps | High intensity, clustered LED type. |
| 11.1.1 | Breaker ON | Red |
| 11.1.2 | Breaker Off | Green |
| 11.1.3 | Spring Charged | Blue |
| 11.1.4 | Auto trip | Amber |
| 11.1.5 | Service position | **Clear** |
| 11.1.6 | **Test position** | **Clear** |
| 11.1.7 | **Trip Circuit Healthy** | **White** |
| **12.0.** | **Switches and push buttons** | Flush mounted on LV compartment door, with shrouded Terminals |
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| 12.1 | TNC Control Switch with pistol grip | Non Lockable spring return to normal position. Shall have mechanical interlock to prevent accidental operation of the switch. Terminals shall be screw type and indelibly marked.<br>2 nos. Contact each for Close and Open, 2 No. NAT,2 nos. NAC and one for normal position contact is required. |
| 12.2. | Local / Remote selector Switch | 2 way, 4 poles in each position lockable and stay put type. |
| 12.3 | Selector switch for voltmeter | 4 Positions ( RY-YB-BR-OFF), 3 ways. |
| 12.4. | Toggle ON/Off switches | For heater / illumination circuit |
| 12.5 | PT Selector switch | 3Position (PT1-PT2-PT3), 8 Poles in each position. Current rating 16 Amps |
| 12.6 | Push button of IED(Relay) Reset | Wired up to specified DI contact of Relay. |
| 12.7 | Emergency trip push button | Red colour with stay put, Current rating 10 Amps. |

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| 12.8 | Push button for Annunciator accept and reset | Two nos Push button, one for annunciator accept and one for annunciator reset to be wired upto annunciator. |
| **13.0** | **Internal wiring** | 1100/750 V grade PVC insulated stranded flexible (FRLS) copper wire. |
| 13.1. | Wire Size | 2.5 sq. mm forCT circuit, 1.5 sq mm for PT and 1.5 sq. mm for control circuit, 4.0 sq. mm for Main, AC & DC and**2.5 sq. mm for others** Bus wiring.<br>· i. A suitable wiring duct system firmly fixed on the panel and having covers shall be installed for front to rear and inter panel wiring to provide easy access for inspection and replacement of the wires.<br>ii. Wiring between terminals of the various devices shall be point to point. Splices or tee connection will not be acceptable.<br>Iii. Facilities for short-circuiting the current transformer secondary while the switchgear is in service shall be provided.<br>iv. Wires shall be suitably bunched adequately supported to prevent sagging. and it shall have sufficient clearance from High voltage system.<br>v. Sufficient Bus wires shall be provided for interconnecting with adjacent units both ways and also for future connection. |
| 13.2 | Wiring Colour code | |
| 13.2.1 | PT and CT | Red Phase : Red, Yellow Phase : Yellow<br>Blue Phase: Blue, Neutral : Black. |
| 13.2.2 | DC Circuit | Gray |
| 13.2.3 | AC Circuit | Black |
| 13.2.4 | Earth | Green |
| 13.3 | Ferrules | At both ends of wire with same stated marking. Plastic ferrules shall be used conforming IS. Interlocked type (one additional red colour ferrule for all wires in trip circuit) |
| 13.3.1 | Ferrule marking | |
| 13.3.1.1 | AC Circuit | H1, H2 ,H3 ............. |
| 13.3.1.2 | Metering CT Circuit | D11, D31, D51 .............. |
| 13.3.1.3 | O/C and E/F protection CT circuit | C11, C31, C51 .................. |
| 13.3.1.4 | REF/Differential protection CT circuit | A11, A31, A51 ......... |
| 13.3.1.5 | Main DC distribution circuit | J1, J2, J3 ............. |
| 13.3.1.6 | Control and protection circuit | K1, K2 ,K3 ................. |
| 13.3.1.7 | Indication and annunciation circuit | L1, L2 ,L3 ................ |
| 13.3.1.8 | Motor circuit | M1, M2 ,M3 ................ |
| 13.3.1.9 | PT circuit | E11, E31, E51, E71............ |

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| 13.3.1.10 | Spare contact circuit | U1, U2, U3 .../ *S1, S2, S3* ....... |
| 13.4 | Lugs | Tinned copper, pre-insulated, fork type and pin type as applicable. Only Ring type is acceptable in CT &PT Circuit. |
| 13.5 | Spare contacts of relays, timers etc | Wired up to the terminal blocks |
| 13.6 | Wiring enclosure | Plastic channels, inter panel wiring through PVC sleeves or suitable grommets. |
| 13.7 | Inter panel wiring | Wiring with ferrule to be terminated in the adjacent shipping section will be supplied with one end terminated and the other end bunched and coiled. |
| **14.0** | **Terminal Blocks (TB)** | |
| 14.1. | Terminal Blocks(TB) | 1100V grade, 10amps min. rating, Nylon 66, screw type suitable for 2 nos. Lead. |
| 14.2. | Terminal for CT & PT secondary leads | With provision for shorting with screw driver operated sliding link.<br>a) CT shorting links shall be provided to short CT circuits under live system condition.<br>b ) Isolation links shall be provided on the trip circuits, alarm and on the VT circuits to allow easy isolation without disconnecting the wires from TBs. |
| 14.3 | Spare terminals | **At least 10% or 4 nos (which is higher)**in each panel |
| 14.4 | TB shrouds & separators | Moulded non-inflammable plastic material |
| 14.5 | Clearance | |
| 14.5.1 | Clearance between 2setsof TB | 100 mm min. |
| 14.5.2 | Clearance with cable gland Plate | 250 mm min |
| 14.5.3 | Clearance between AC / DC set of TB | 100 mm min |
| **15.0** | **Test Terminal blocks for energy meter** | 13 Ways, 50A, 1100V with back connection. Dia of Entries is 5 mm & overall dimension 250X50 mm, covered with insulated cover. TTB should be suitable for 3 Phase 4 Wire Energy Meter.<br>TTB used for CT Circuit should be Link type. Screw type is not acceptable. |
| **16.0** | **HT Cable Type** | 3 Core XLPE,11 kV grade , Aluminium,300mm$^2$ |
| **17.0** | **HT Cable Termination** | Cable entry from rear bottom side for Incomer and Outgoing feeders .The cable termination shall be located at least 250 mm from the CT primary terminals. For Incomer, double cable termination arrangement to be provided with two sets of nut and bolts. Copper terminator strip of adequate size shall be provided for termination of cables and shall have adequate height inside to accommodate the heat shrinkable type indoor cable termination. |

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| **18.0** | **Multi-core cable termination box-- Purpose** | Incoming Panel- External DC supply from DCDB, External AC supply from ACDB, Remote Electrical Close and trip operation, Inter trip operation from upstream 33 kV circuit, Breaker spare aux. Contacts. Spare contacts of L/R Switch and Spring Charge Limit Switch etc.<br><br>Feeder Panel -- Remote electrical close/trip operation, remote indication for close, trip , spring charged and trip ckt supervision, status of L/R switch, Local annunciation for remote close and trip operation, Breaker spare aux. Contacts.<br><br>Bus Coupler -- External DC supply from DCDB, External AC supply from ACDB, Remote Electrical Close and trip operation, Breaker spare aux. Contact.<br><br>External DC and AC supply arrangement to be provided in Incomer and Bus Coupler panel only in a switch board. |
| **19.0** | **Operation Counter** | Each breaker shall be provided with operation counter. |
| **20.0** | **Vacuum Interrupter** | Capable to withstand minimum 100 full short circuit operations at 26.3 kA as per test duty 1 to 5 of IEC-56. The continuous Current rating should be more than 800A. Manufacturer's test report/literature to be supplied along with tender. |
| **21.0** | **Insulated spacer** | It shall be made of insulated fibre glass materials shall be provided in High voltage system, to protect from absorption of moisture present in the air. |
| **22.0** | **PT selection scheme** | Each Incoming and outgoing breaker panel shall be provided with suitable manually operated selector switch having break before make type contacts including necessary wiring for selection of voltage from different PT secondary's available for connection with energy meters **and numerical relay for Incomer panel. One output of the numerical relay to be configured as "PT under voltage" for one of the annunciator window as Non Trip indication.**<br><br>Each panel should be provided with 3 PT scheme and current rating of the PT Selector Switch should be minimum 16 Amp.<br>Voltmeter of incoming breaker panel shall be connected directly from respective PT secondary of associated unit. |
| **23.0** | **Relays for protection and control** | |
| 23.1 | General Features | |
| 23.1.1 | Relay type | Numerical with self monitoring features |
| 23.1.2 | Mounting | Flush mounted IP 5X with key pad on front |



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| 23.1.3 | Relay characteristics | Numerical IED with Multiple characteristics like IDMT, DMT, instantaneous with compatible choices of time delays and multiple settings for multiple functions like overcurrent, earth fault etc. along with control of breaker, measurement and status etc. As per detailed technical specification attached in a separate sheet (Annexure-E). |
| 23.1.4 | Relay communication | With IEC 61850 protocol by native two nos. RJ45 /F.O, 100 MBPS communication ports. Separate front port (USB/RJ45) for Relay parameterisation. |
| 23.1.5 | Relay input signal | From CT, with auxiliary DC supply |
| 23.1.6 | Relay terminals | Shall be screw type terminals large enough to accommodate 2.5 sq. mm. cables and shall be located at the back of the Relay. |
| 23.1.7 | Relay contacts | Shall make firmly without bounce and the relay mechanism shall not be affected by panel vibration or external magnetic field. |
| 23.1.8 | Electromagnetic Compatibility | Relay thermal rating shall be such that the fault clearance times on any combination of current and time multiplier settings shall not exceed the thermal withstand capability of the relay. Compliant to EMC directives as per IEC. |
| 23.1.9 | Relays for auxiliary, supervision, trip and timer relays | Static or attracted armature type with short pickup time of less than 30 ms. |
| 23.1.10 | Relay reset | Self reset contacts except for Master Trip relays. |
| 23.1.11 | Operation indicators | With hand reset operation indicators (flags) or LED swith Push buttons for resetting, for analyzing the cause of operation |
| 23.2 | Master Trip relay | High Impedance and High speed relay flush mounted having coil cut-off contact with at least 4NO and 4NC contact and **electrical reset facility** capable to make, carry and break trip coil current of Circuit Breaker and capable for future integration with SCADA. It should be immune to capacitance discharge currents and leakage current. Operating time should be less than 20ms. Terminals shall be screw type to accommodate 1.5 sq. mm. Cable and located at the back of the relay. Terminals shall be clearly marked. Contact configuration shall be drawn on the relay casing. |
| 23.3 | Fault recording | Relay shall have the facility for recording of various parameters during a fault. It should be possible to set the duration of record through settable pre fault and post fault time. It should be possible to down load the data locally or from SCADA remotely. |
| 23.4 | Auxiliary supply | Operate on available 30 V. DC supply. To reduce the effect of Electrolysis, relay coil shall be so connected such that they are not continuously connected from the positive pole of the station battery. |
| 23.5 | Test facility | In built. |



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| 23.6 | Protection in Incomer & Outgoing panel | 3 phase overcurrent protection & Earth fault protection |
| 23.7 | **Additional requirement** | |
| 23.7.1 | For each breaker panel | To be provided with anti pumping relay/contactor (94). (plug-in type contactor will not be accepted) |
| 23.7.2 | Auxiliary relays, coupling relays, transducers etc. | To effect interlocks and to exchange signals of status & Control from remote. |
| **24.0** | **Additional Requirement** | The schematic diagram of control and indication circuit on a durable sticker shall be fixed in a suitable place in Incomer and feeder panels and Annunciation scheme in the Bus Coupler panel.<br>The wiring arrangement, position of fuse, TBs etc, in the panel should be such that each and every connection can be accessed easily. Fuse tops can be opened /inserted easily.<br>Only ring type sockets should be used for wire termination in TB/ Relays/ Meters/TTB etc. Fork type socket can be used in Annunciator only.<br>Separate switch to be used for cubicle light instead of Door switch.<br>Circuit Label incorporating identification, information of incoming/outgoing/bus coupler breaker shall be provided.<br>Connection diagram plates for the PT and CT shall be provided as per provision in the relevant IS.<br>All name, rating , circuit label, connection diagram plate shall be fixed with screws or by riveting instead of fixing by adhesive. |
| **25.0** | **Annunciation Scheme** | 6Window shall be provided in each incomer and outgoing panel, 8 window for Bus Coupler panel. It should be Microprocessor based having inbuilt Accept/Reset/Test/Mute push buttons for trip and non-trip functions. DC operated common Hooters shall be provided in Bus Coupler breaker panel. It shall have provision of inbuilt watch Dog and Fast fault indication with Trip (O/C , E/F and other trip signals if any )and non-trip alarm i.e AC supply fail, main DC supply fail at Bus Coupler. The functional details and scheme will be decided during finalisation of drawings. |
| **26.0** | **;pace Heaters** | |
| 26.1 | Space heaters | 60 W, 230 V AC each with timer controlled and switch for isolation. A Thermostat control unit with variable temperature control setting shall be installed to control the heater. The 240 V AC supply for the heater shall be controlled by a suitably rated single pole miniature circuit breaker. |
| 26.2 | Space heater location | Two nos. each in Breaker & HV cable compartment to be mounted on an insulator. |



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| 26.3 | Space heater Monitoring | One AC Ammeter with 0-1.0Amp range shall be provided in series with the heaters to monitor the current drawal of the Heaters. Size :96x48mm. |
| **27.0** | **Switch and sockets** | |
| 27.1. | Illumination Lamp | For LV ,cable chamber & energy meter |
| 27.2. | 240V AC, 15A /5A Switch & socket | In LV chamber |
| **28.0** | **Name Plates and rating plate of switchgear** | |
| 28.1 | Material | Material shall be stainless steel/anodized aluminium and shall not be deformed under the service condition. The entries shall be indelibly marked by engraving with black letter on white background or vice-versa as specified and 3mm thick and 100mmX150mm(approx). |
| 28.2 | Lettering | Engraved, with white letters on black background, for use inside the control compartment, a white label, engraved in Black Letters and numbers shall be used. Each major equipment shall be provided with a rating plate containing the required information as specified in the relevant IEC standard. |
| 28.3 | Name plate for feeder description & number | On front and rear side of each panel giving feeder description. On the rear side the name plate shall be fixed on the back cover. |
| 28.4 | Identification plates/labels at each cubicle and each instrument | Approved design (not stickers) |
| 28.5 | Manufacturer name plate | On front top door of panel |
| **29.0** | **Rating Plate of Switchgear** | Name of the Manufacturer and Month/Year of Manufacture |
| | | P.O No. and Date |
| | | Type, Designation and Serial No. |
| | | Rated Voltage and Current |
| | | Lightning impulse withstand voltage |
| | | 1 minute Power frequency withstand voltage |
| | | Rated frequency |
| | | Rated Current |
| | | Breaking Capacity |
| | | Short time current making capacity |
| | | Operating sequence |
| | | Rated voltage of closing and opening coil |
| | | Rated voltage of spring charging motor |



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| | | Make and type of VI<br>Property Label "Property of WBSEDCL"<br>Guarantee for Five Years |
| **30.0** | **General** | a) Materials shall be new; the best quality of their respective kinds and such as are usual and suitable for work of like character. All materials shall comply with the latest issues of the specified standard unless otherwise specified or permitted by WBSEDCL.<br>b) Workmanship shall be of the highest class throughout to ensure reliable and vibrations free operations. The design, dimensions and materials of all parts shall be such that the stresses to which they may be subjected shall not cause distortion, undue wear, or damage under the most severe conditions encountered in service.<br>c) All parts shall conform to the dimensions shown and shall be built in accordance with approved drawings. All joints, datum surfaces and meeting components shall be machined and all castings shall be spot faced for nuts. All machined finishes shall be shown on the drawings. All screw, bolts, studs and nuts and threads for pipe shall conform to the latest standards of the International Organization for Standardization covering these components and shall all conform to the standards for metric sizes.<br>d) All materials and works that have cracks, flaws or other defects or inferior workmanship will be rejected WBSEDCL.<br>e) Each of the Switchgear panel shall be of unitised construction with all necessary accessories like end covers etc. However the design shall allow for extension on both sides without limit. Busbar design shall be such that panel to panel interconnection can be carried out without difficulty as and when required. **In case of Single feeder panels, Identical Bus Bars for connection with adjacent Incomer / Bus Coupler / Feeder needs to be supplied.**<br>f) Explosion vents of suitable design shall be provided on the roof sheet of the busbar/cable/CT's chambers so as to enable discharge of explosive gases from inside during a flashover.<br><br>g) All primary current carrying conductor bar shall be CU made and minimum current carrying capacity shall be as per CB rating if otherwise not specified. Current density to be considered not more than 1.6 A/sqmm. |
| 30.1 | Assembly | Necessary items of equipment shall be assembled in the factory prior to shipment and routine tests shall be performed by the manufacturer as per the requirements of the latest issue of IEC as specified under each equipment in these specifications to demonstrate to the satisfaction of the WBSEDCL that the switchgear panels comply with the requirements of the relevant IEC standards. |

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| 30.2 | Casting | Casting shall be true to pattern, of workmanlike finish and of uniform quality and condition, free from blowholes, porosity, hard spots, shrinkage defects, cracks or other injurious defects, shall be satisfactorily cleaned for their intended purpose. |
| 30.3 | Welding | Wherever welding is specified or permitted, a welding process, including stress relieve treatment as required if necessary, conforming to an appropriate and widely recognized professional standard shall be used. All welders and welding operators shall be fully qualified by such a standard. |
| 30.4 | Operational Details | Instructions shall be engraved on the switchgear panel, on the circuit breaker compartment describing in simple steps how to carry out correct and safe isolation, racking-in and racking-out switching operations on the circuit breaker. Similar details should be provided for the operation of the earth switch. |
| **31.0** | **Surface treatment & painting** | |
| 31.1 | Surface treatment | Sand blasting or by seven tank process |
| 31.2 | Paint type | Powder coated. Pure polyester base grade A structure finish |
| 31.3 | Paint shade | RAL 7032 for external & internal surface |
| 31.4 | Paint thickness | Minimum 60microns |
| **32.0** | **Type Test** | |
| 32.1 | Type Tests | **The equipment offered in the tender should have been successfully type tested at NABL accredited third party laboratories in India or equivalent STL accredited international laboratories in line with the relevant IEC/Indian standards and technical specification,**<br>**The bidder shall be required to submit complete set of the type test reports along with the offer.** |
| 32.2 | Type test report validity period | **Validity of type test shall be as per latest CEA guidelines, i.e., 5 (five) years from the date of offer.** |

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| 32.3 | Type test report to be submitted along with the bid | **A . Switchgear Panel (with circuit breaker installed)**<br>a. Lightning Impulse Voltage withstand Test<br>b. H.V. dry 1 min power frequency withstand test<br>c. Short time and peak withstand current test<br>d. Short circuit test with basic duties<br>**e . Single phase breaking capacity test.**<br>**f. Capacitive current switching test confirming to relevant IEC:-**<br>**i) Line charging breaking current test: 10A**<br>**ii) Cable charging breaking current test:25A**<br>**iii) Single capacitor bank breaking current test: 400A**<br>g. Temperature Rise test<br>h. IP Test<br>i. Internal Arc Test as per IEC 62271-200<br>j. Horizontal Acceleration due to seismic force (if available)<br>**B. Circuit Breaker**<br>a. Mechanical Endurance Test as per M2 Class of IEC<br>**b. Class E2 circuit-breakers intended for use without auto-reclosing duty as per clause no 6.112.1 of IEC. Basic short-circuit test series shall consist of test-duties T10, T30, T60, T100s and T100a as per clause no 6.106 of IEC and all sequence to be done without any maintenance in between the test and test reports to be made in single occurrence.**<br><br>**C. Current Transformer**<br>a. Short Time Current Test<br>b. Impulse Voltage Withstand Test<br>c. Temperature Rise Test<br>**D. Potential Transformer**<br>a. Impulse Voltage Withstand Test<br>b. Temperature Rise Test<br>**Copies of test certificates in respect of following bought out items:-**<br>a. Vacuum Interrupter.<br>b. Insulators<br>c. Bus Bar Material<br>d. Terminal connectors<br>e. KEMA / CPRI Certification for relay/IED i.r.o. IEC61850 compliance<br><br>**Note : All the type test report on Switchgear Panel**<br>**& Circuit Breaker to be conducted with offered Vacuum Interrupter.** |

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| 33.0 | Accessories | **All loose materials (Accessories) shall be supplied inside the cable compartment of each breaker panel with properly packed. List of loose materials (Accessories) are given in Table-1.**<br>**List of loose materials shall also to be provided with the packing.**<br><br>**Each panel shall be supplied with inter panel BUS wire with ferrule marking irrespective of the position of the panel.** |
| 34.0 | **Type Test after issuance of Order** | Besides submission of Type Test Report, carried out within five years as per Tender Specification, Type Test at the discretion of WBSEDCL, shall have to be arranged by the successful Manufacturer from any lot offered for inspection, sample chosen at random after successful Routine Test by our Inspection Team, as per relevant ISS from CPRI/NABL accredited/Government recognized Test House or Laboratory in presence of WBSEDCL'S representative.<br>**However the necessary cost of the Type Test charges will be reimbursed to the party on production of necessary supporting documents.** |
| 35.0 | **Equipment Device Schedule** | Enclosed in Annexure-D. Bidder shall have to fill up GTP as per Annexure-G, Annexure-H, Annexure-J, Schedule-A, Schedule-B, Schedule-C. Bidder shall have to submit undertakings from Relay and Meter manufacturer as per Schedule-I and II. |



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| 36.0 | **Drawing and documentation** | Bidder shall submit following drawing/documents along with the bid:-<br>1. GA of indoor **requisite set of** panel Switchgear showing constructional features and space required in the front for withdrawal of breaker truck and in back, other accessories, power and control cable entry with plan elevation and views.<br>2. Sectional view of incomer, bus coupler & feeder panels with parts list.<br>3. GA of Circuit Breaker truck.<br>4. GA of Current Transformer<br>5. GA of Potential Transformer.<br>6. Technical particulars of Switchgears and brochures for technical data sheet of vacuum interrupter<br>7. Relay and device list (Bill of Quantity) with complete details.<br>8. Technical manual for Relays, Meters, Switches, Instrument transformer, Terminal block and Test Terminal Block.<br>9. Four copies of drawing, data and manuals containing above shall be submitted for approval and afterwards for final distribution.<br>10. Two sets of manual, leaflet and drawing for multi panel board shall be submitted separately to the C.E(Testing), Distribution Testing Deptt., WBSEDCL.<br><br>**Successful tenderer shall furnish all above drawings and following additional drawings for approval before commencement of supply**:-<br>i. Foundation details for requisite set of Panel Switchgear.<br>ii. Equipment door layout for incomer, bus coupler & feeder panels.<br>iii. Schematic Diagram for incomer bus coupler & feeder section of Switchgear<br>iv. Protection Circuit, Metering circuit, DC control circuit for incomer bus coupler & feeder section of Switchgear<br>v. Annunciator and Alarm scheme.<br>vi. P.T. supply change over scheme.<br>ix. Terminal block details for incomer, bus coupler & feeder section.<br>vii. Cross section view for CTs.<br>viii. Name Plate & Connection diagram for CTs.<br>ix. Cross section view for PTs.<br>x. Name Plate & Connection diagram for PTs.<br>xi. Rating Plate details of the Panel<br>xii. Manual for installation, operation and maintenance procedure. |

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| **37.0** | **Guarantee** | Guarantee of the total equipment including any integral part of the equipment should be for a period of 5 (five) years from the date of last despatch. |
| **38.0**<br><br>38.1 | **Routine & Acceptance Test** | All the switchgear panels shall be tested in accordance with the requirement of IEC 62271-200.<br>Tests shall be carried out on the circuit breakers as per the requirement of IEC 62271-100.<br>Current transformers and Voltage transformers shall be tested in accordance with the requirement of IEC 60044-1, IS 2705 and IEC 6044-2, IS 3156 respectively. The Protection Relay shall be tested in accordance with IEC 60255 & Energy Meter shall be tested as per relevant IS.<br>All routine tests at manufacturer's works shall be carried out and Test Reports are to be submitted to CE, P&C, WBSEDCL.<br><br>All Acceptance tests shall be carried out at manufacturer's works on every lot offered for inspection as per relevant IS & IEC in presence of the WBSEDCL's representatives. Selection of samples for acceptance test as well as rejection and retesting shall be guided by relevant IS. The entire cost of acceptance and routine test that to be carried out shall be treated as included in the quoted price of tendered items.<br><br>Six copies of test reports duly signed by the inspecting officers, shall be submitted to the Chief Engineer, P&C Department , Bidyut Bhavan (4th floor) Salt Lake, Kol -91. |
| 38.2 | Complete Switchgear Panel test | a ) Dimensional Checks<br>b) Operational Tests<br>c) Primary Injection Tests<br>d) Calibration Tests on Relays and Instruments<br>e) Power frequency Withstand Test<br>f) Insulation Test<br>g) Contact resistance test of Primary joints<br>h) Power frequency Withstand Test on secondary Wiring |
| 38.3 | Circuit Breaker test<br>( six copies of test report to be submitted with Offer) | Routine tests.<br>a) Operation test.<br>b) Dielectric Test on the main circuit.<br>c) Dielectric tests on controls and auxiliary circuits.<br>d) Measurement of resistance of the main circuit.<br>e) Timing test<br>f) Design and visual checks<br>g) Test Certificate for all resin cast/moulded components of circuit breakers whose partial discharge measurement are specified in the relevant IEC (No. 62271-100, Clause No.6.2.9) |

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| 38.4 | Current Transformer Test ( six copies of test report to be submitted with Offer) | Routine tests shall be carried out at the manufacturer's plant as per the requirement of IEC 60044-1, **IS 16227 (PART 1 & 2)**as listed below:<br>i. Polarity test and verification of terminal markings test<br>ii. Ratio and phase angle error test(accuracy class; composite error test)<br>iii. Power frequency tests on primary and secondary windings<br>iv. Power frequency withstand tests between sections(windings)<br>v. Inter-turn over voltage tests<br>vi. Partial discharge measurement |
| 38.5 | Voltage Transformer test ( six copies of test report to be submitted with Offer) | Routine tests shall be carried out at the manufacturer's plant as per the requirement of IEC 60044-2, **IS 16227 (PART 1 & 3)**as listed below:-<br>(a) Polarity tests and verification of terminals<br>(b) Power frequency withstand tests on Primary windings<br>(c) Power frequency withstand tests on secondary windings<br>(d) Power frequency withstand tests between sections<br>(e) Determination of errors<br>(f) Partial discharges measurement |
| | Protection Relay Tests ( six copies of test report to be submitted before with Offer) | a. Relay Pick-up test for all functions and phases<br>b. Relay timing test for all functions and phases<br>c. Conformance Test as per IEC 61850 |

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# Table-1

| LIST OF LOOSE MATERIAL | | | | |
|---|---|---|---|---|
| SI.NO | Object description | INCOMER & | OUTGOING PANEL | BUS COUPLER PANEL |
| | | Qty | Qty | Qty |
| 1 | BUS BAR ( AS PER DESIGN & APPROVED DRAWINGS) | 1 SET | 1 SET | 1 SET |
| 2 | RACK IN/RACK OUT HANDLE | 1 NO | 1 NO | 1 NO |
| 3 | PANEL COUPLING HARDWARE KIT | 1 SET | 1 SET | 1 SET |
| 4 | HT HARDWARE FOR BUS BAR | 1 SET | 1 SET | 1 SET |
| 5 | FOUNDATION BOLT | AS PER DESIGN & APPROVED DRAWINGS | AS PER DESIGN & APPROVED DRAWINGS | AS PER DESIGN & APPROVED DRAWINGS |
| 6 | BUS BAR SHROUDS | AS PER DESIGN & APPROVED DRAWINGS | AS PER DESIGN & APPROVED DRAWINGS | AS PER DESIGN & APPROVED DRAWINGS |
| 7 | END COVER | 1 | - | - |
| 8 | END PANEL GROMMET | 1 | - | - |
| 9 | SHORTING EARTHBAR | 1 | - | 1 |



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## ANNEXURE –A

## MANDATORY TECHNICAL PARTICULARS

| | | |
|---|---|---|
| **1.0** | **Switchgear** | |
| 1.1 | Type | Metal clad, air insulated with VCB type circuit Breaker |
| 1.2 | Service | Indoor |
| 1.3 | Mounting | Freestanding, floor mounted |
| 1.4 | System voltage | 11 kV |
| 1.5 | Highest System voltage | 12 kV |
| 1.6 | Frequency | 50HZ, ± 3% |
| 1.7 | Phase | 3 |
| 1.8 | Ratedcurrent @ 50 DEGC ambient | 800 Amp. |
| 1.9 | Short time rating for 3 sec. | 18.4 kA |
| 1.10 | Insulation level (PF rms / impulse peak) | 28 / 75 kV |
| 1.11 | System ground | Solidly earthed |
| 1.12 | Enclosure degree of protection | IP 5X for high and low voltage compartment |
| 1.13 | Bus bar – Main @ 50 DEGC ambient | 1600A, Current Density 1.6 Amp/sq.mm.(Max) |
| 1.14 | Material | electrolytic copper, tinned /Silver plated |
| 1.15 | Bus Bar sleeve | Full voltage sleeved with **shrouds** on joints |
| 1.16 | Bus identification | Colour coded (R-Y-B) |
| 1.17 | Bus end connection | To be capable to safely withstand stress due to max. Short circuit current and thermal expansion. Necessary provision to be made for testing current transformer primary by removing insulated portion without difficulty. |
| 1.18 | Temperature rise above Ambient | 40DEGC for conventional joints, 55DEGC for silver plated joints |
| 1.19 | Riser Bus | electrolytic copper, tinned /Silver plated,<br>800A, Current Density 1.6 Amp/sq.mm.(Max) |
| 1.20 | For the design and erection of Busbars the following minimum clearance shall be observed | |
| 1.21 | Phase to Phase(mm)<br>(without Insulating Sleeve) | 110 mm (approx) |
| 1.21.1 | Phase to earth(mm)<br>(without Insulating Sleeve) | 90 mm(approx) |
| 1.21.2 | Minimum nominal creepage distance | 19mm/kV. |
| 1.22 | Auxiliary DC Supply | 30V DC shall be controlled by suitably rated miniature circuit breaker. |



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| | | |
|---|---|---|
| 1.23 | Auxiliary AC supply | 240V AC 50 Hz shall be controlled by suitably rated miniature circuit breaker. |
| 1.24 | Hardware | GI |
| 1.25 | Earth bus | Electrolytic Copper, Rectangular shape, 25X6mm(Min) and to carry 18.4 KA for 1 Second. |
| 1.26 | Power and control cable entry | From bottom for Power Cable. For control cables through Remote Terminal Box installed at the backside of the Panel. |
| **2.0** | **Circuit Breaker** | |
| 2.1 | Type of isolation & Draw out | Horizontal isolation & Horizontal Draw out. |
| 2.2 | Voltage class, insulation level, short time rating | As specified for switchgear |
| 2..3 | Rated current | 800 Amps. |
| 2.4 | Duty cycle | O – 0.3sec –CO – 3min –CO |
| 2.5 | Short circuit rating | |
| 2.5.1 | AC sym. Breaking current | 26.3 KA (for Vacuum Interrupter) |
| 2.5.2 | Short circuit making current | 65.75KA (for Vacuum Interrupter) |
| 2.6 | Operating time | |
| 2.6.1 | Break time | Less than 80 ms. |
| 2.6.2 | Make time | Less than 100 ms. |
| 2.7. | Range of auxiliary voltage | |
| 2.7.1 | Closing | 85% - 110% |
| 2.7.2 | Tripping | 70% - 110% |
| 2.7.3 | Spring charging | 85% - 110% |
| 2.8 | Auxiliary Switch | Properly rated and robust in nature shall be provided in all panels and wired upto SIC (Secondary Isolated Contact) with identical SIC and ferrule numbers and sufficient number of NO and NC contacts for the following:-<br>CB 'ON' for Local indication –NO contact<br>CB'ON' for Remote indication—NO contact<br>CB'OFF' for Local Indication --- NC Contact<br>CB'OFF' for Remote Indication – NC Contact<br>CB 'Auto Trip' for Local and Remote Indication—NC Contact. Other Contacts used as per scheme for closing and tripping and trip ckt supervision. |
| 2.8.1 | No. of spare contacts in Aux. Switch for WBSEDCL use | Minimum 6 NO + 6 NC |
| 2.9 | No. of spare contacts of service and test position limit switch contact | 2 NO+2NC |
| 2.10 | No. of spare contacts of Spring charge limit switch | 2 NO + 2 NC |
| **3.0** | **Current Transformers for Incoming Panel** | |
| 3.1. | Voltage class, insulation level | As specified for switchgear |



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| | | |
|---|---|---|
| 3.2. | Type | Triple core, Dual ratio, Cast resin, Indoor type . Marking on primary and secondary terminal as per requirement of IEC 60044-1, IS 16227 (PART 1 & 2)and shall be indelibly marked. The secondary terminals shall have screw type terminals. |
| 3.3. | Class of insulation | Class E or better |
| 3.4. | Ratio | 600/400/1-1-1A |
| 3.5. | Number of secondary | Three, Core 1 : Metering, Core 2 :Protection , Core 3 : Protection |
| 3.6. | Accuracy class | Core 1 : 0.5, Core 2 : 5P15, Core 3 : PS. |
| 3.7 | ISF of Metering Core | <=5 at lower ratio |
| 3.8 | ALF of Protection Core | 15 |
| 3.9 | VA output | 10VA for both protection and metering |
| 3.10 | | |
| 3.11 | Excitation current and other particulars of PS class CT's | 30mA at Vk/2 (at lower ratio), $V_k$>=170 V at LR |
| 3.12 | Other particulars | $R_{CT} \leq 4$ ohms at 75° C at higher ratio, Panels to be supplied with CT ratio set at higher ratio. Common terminals for CT secondary shall be shorted and earthed. Only ring type socket to be used in CT circuit.<br>P1 terminal of CT shall be at Bus side. |
| 3.13 | Short time current rating | 18.4 kA for 1 second |
| **4.0** | **Current Transformers for Outgoing Panel** | |
| 4.1. | Voltage class, insulation level | As specified for switchgear |
| 4.2. | Type | Dual core, Dual ratio, Cast resin, Indoor type. Marking on primary and secondary terminal as per requirement of IEC 60044-1, IS 16227 (PART 1 & 2)and shall be indelibly marked. The secondary terminals shall have screw type terminals. |
| 4.3. | Class of insulation | Class E or better |
| 4.4. | Ratio | 200-100/1-1A |
| 4.5. | Number of secondary | 2, Core 1 : Metering, Core 2 :Protection |
| 4.6. | Accuracy class | Core 1 : 0.5, Core 2 : 5P15. |
| 4.7. | ISF of Metering Core | <=5 at lower ratio |
| 4.8. | ALF of Protection Core | 15. |
| 4.9 | VA output | 10VA for both protection and metering |
| 4.10 | Short time current rating | 18.4 kA for 1 seconds |

| | | |
|---|---|---|
| 4.11 | Name Plate of CT indelibly marked | Name plate for the current transformer shall be provided with all the required details as per IEC 60044-1, IS 2705 Standards, including :-<br>i) The manufacturer's name<br>ii) Serial number and a type designation<br>iii) Rated primary and secondary current<br>iv) Rated frequency<br>v) Rated output and corresponding accuracy class for each secondary winding, including the rated accuracy limit factor and Instrument security factor for protection and metering secondary windings respectively.<br>vi) The highest voltage of the equipment<br>vii) The rated Insulation level<br>viii) The rated short-time thermal current (***Ith***) and the rated dynamic current<br>ix) Class of Insulation<br>x) Rated continuous thermal current<br>xi) Property Label – " Property of WBSEDCL"<br>xii ) Guaranteed for 5 Years |
| **5.0** | **Voltage Transformers** | |
| 5.1. | Type | Dry / Resin cast, draw out type, 3-phase, 5-limb or it should be combination of three no Single Phase PT housed in a withdrawable carriage complete with HV fuses. Copies of Type test certificate and routine test certificate as per IEC 60044-2, IS 16227 (PART 1 & 3)shall be submitted with the tender for evaluation purpose. Each PT shall be provided with Vector diagram plate as per relevant provision of IS showing primary and secondary terminals. |
| 5.2. | Rated Voltage | |
| 5.2.1 | Primary | 11000 /√3V |
| 5.2.2 | Secondary | 110 /√3V |
| 5.3 | No of phases | 3 |
| 5.4 | No. of secondary windings | 1(one) |
| 5.5 | Method of connection | Star/Star with both neutral earthed |
| 5.6 | Rated voltage factor | 1.2 continuous, 1.9 for 8 hours. |
| 5.7 | Class of insulation | Class E or better |
| 5.8 | Accuracy class | 0.5 |
| | | |
| 5.9 | Primary and secondary fuses | HRC current limiting type, primary fuse Replacement shall be possible with VT in withdrawn position |
| 5.10 | Voltage class | 12 kV |
| 5.11 | VA Burden | 100VA per phase |

| | | |
|---|---|---|
| 5.12 | PT mounting | Rail mounted on top of the Incomer Panel and Primary connected on bus side. It can be plugged into and withdraw from service by pulling or pushing the PT by the handle provided on the PT. This action traverses the PT along the rails and shall automatically operate the spout shutters. The shutter drive also forms a latch which holds the PT in the service position and this latch shall be required to be released before PT can be isolated. |
| 5.13 | PT secondary circuit | The secondary circuits from all the PTs of the incoming panels shall be wired up to all the outgoing panels so that supply to the potential coils of the meters can be maintained from any of the PTs. Plug and socket type contacts shall be provided in the PT secondary. PT secondary terminals shall be inaccessible from outside. Necessary provision shall be made for sealing PT while in racked in position. Vector of the PT will be Both Y/Y connection with both neutral earthed. |
| 5.14 | Name Plate of PT indelibly marked | Name plate for the voltage transformer shall be provided with all the required details as per IEC 60044-2, IS 16227 (PART 1 & 3)Standards, including :-<br>i) The manufacturer's name<br>ii) Serial number and a type designation<br>iii) Rated primary and secondary voltage<br>iv) Rated frequency<br>v) Rated output and corresponding accuracy class for each secondary winding,<br>vi) The highest voltage of the equipment<br>vii) The rated Insulation level<br>ix) Class of Insulation for the equipment<br>x) Rated voltage factor and corresponding rated time.<br>xi) The use each secondary winding and it's corresponding terminals<br>xii)Property Label – " Property of WBSEDCL"<br>xiii ) Guranteed for 5 Years |
| **6.0** | **Monitoring from Remote** | Suitable arrangement to be provided to monitor the following conditions of 11kV VCB at SCADA Control Centre end through installing RTU by WBSEDCL:-<br>1 . DC healthy<br>2 .Breaker is 'OFF'<br>3 .Breaker is 'ON'<br>4 .Spring for closing mechanism is charged<br>5 .Breaker is in Local<br>6 .Breaker is in Remote<br>7. Trip Circuit healthy<br>8 .Tripping due to O/C<br>9 .Tripping due to E/F |



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| | | |
|---|---|---|
| **7.0** | **Closing from Remote** | Suitable arrangement should be provided to close the breaker from remote end in case following conditions are satisfied:-<br>1. DC healthy<br>2. Breaker is 'OFF'<br>3. Spring for closing mechanism is charged<br>4. Breaker is in Remote<br>5. Trip Circuit healthy<br>6. Breaker is not anti-pumping |
| **8.0** | **Tripping from Remote** | Suitable arrangement should be provided to trip the breaker from remote end in case the breaker is ON and certain preset conditions to be satisfied which will be decided during finalisation of drawings. |

**Annexure - B**
**Standard Make of Relay and Fitments**

| | |
|---|---|
| VCB | Panel Manufacturer should be the VCB Manufacturer. |
| Vacuum Interrupter | CGL / BEL / SCHNEIDER / SIEMENS / ABB |
| Relays | Alstom, ABB, Siemens, Schneider, GE or any other reputed relay manufacturers fulfilling relays specification as per Annexure-E of the technical specification |
| Current Transformer | Plastofab, BMC, Pragati, ECS, Kappa. |
| Potential Transformer | Plastofab, BMC, Pragati, Audio-Vision, ECS. |
| Breaker Control Switch/LR Switch | Kaycee / Alstom / Recom/Switron/L&T/Siemens/ABB/GE |
| Voltmeter/PT Selector switch | Kaycee / Recom/Switron |
| Digital Ammeter / voltmeter and Analog DC Volt Meter | AE / Rishav / Secure/Accord/Schneider/Meco |
| Energy meter | Secure Meter Ltd. / Larsen & Toubro Ltd. / Landis & Gyr Ltd. / Genus Power Infrastructure Ltd. |
| Push Buttons | Alstom / Kaycee / Vaishno/Teknic/Lumen/STS/ Siemens/L&T/Schneider/JVS/GE/ESBEE |
| Indicating Lamps with lenses | Alstom / Kaycee / Vaishno/Teknic/Lumen/STS./ Switron/JVS/GE/ESBEE |
| Panel wiring | ECKO / PHOENIX/Finolex/Havels/KEI/RR Kables/Poly Cab (with ISI mark) |
| Hooter/Buzzer/Bell | ANCHOR / KAYCEE/ Vaishno/STS/JVS/Bharani/Alan/EAPL |
| Annunciator | MINILEC / ALAN/INSTALARM/EAPL/ |
| MCB | Siemens/L&T/Schneider/ABB/Kaycee/Legrand/GE/EATON/C&S |
| Terminal Block | Elmex/Connectwell/Phoenix / Wago |
| TTB | DAV Industries/CGL/JVS/Unitech Sales/Keizen/Plastofab/Deepl |

# Annexure-C

**Legend of Devices associated with Multi panel Shunt trip vacuum Circuit Breaker**

| Symbol Reference | Description |
|---|---|
| A1, A2, A3 & AH(heater) | Ammeter |
| V and $V_{DC}$ forBus coupler | Voltmeter |
| VS | Manual Voltmeter Selector Switch |
| EM | Energy Meter |
| CS | Control switch T-A / T-N-A / C-C spring return type |
| IL-R | CB 'ON' Indication Red lamp |
| IL-G | CB 'OFF' Indication Green lamp |
| IL-W | 'Trip /Close signal received from Remote Indication white lamp |
| IL-B | "Spring charged" Indication Blue lamp |
| IL-A | CB " Auto trip" Indication Amber lamp |
| PB | Push Button |
| ANN | DC operated electric Annunciator |
| H,HS,TH | Heater, Heater Switch, Thermostat |
| FS | Fuse |
| LK | Link |
| MCB1 | MCB, 2 pole 16 A for DC supply |
| MCB2 | MCB, 2 pole 16 A for AC supply |
| MCB3 | MCB, 2 pole for spring charging motor supply |
| MVS | Manual PT selector switch |
| IR-I | Remote inter tripping contact from 33 kV Transformer Control and relay Panel |
| TC | Tripping Coil |
| CC | Closing Coil |
| APR | Anti-pumping Relay |
| 86 | Tripping Relay |
| 52 | Vacuum Circuit breaker |
| 52a,52b | NO and NC contacts of Breaker Auxiliary switch respectively |
| PT | Potential Transformer |
| CT | Current Transformer |
| TTB | Test Terminal Block |
| 51/50 R-Y-B-N | O/C and E/F Relay |

## Annexure D

| Sl no. | Device | Quantity | | |
|---|---|---|---|---|
| | | Incomer | Feeder | Bus coupler |
| 1. | a) Numerical O/C & E/F Relay(Directional) | 1 No. | NIL | NIL |
| | b) Numerical O/C & E/F Relay | | 1 No. | 1 No. |
| 2. | Trip Relay | 1 No. | 1 No. | 1 No. |
| 3. | AC Supervision Relay | Nil | Nil | 1 No. |
| 4. | Annunciator | 1 No. | 1 No. | 1 No. |
| 5. | AC Ammeter ( CT operated ) | 3 No. | 3 No. | Nil |
| 6. | AC Voltmeter | 1 No. | Nil | Nil |
| 7. | DC Voltmeter | Nil | Nil | 1 No. |
| 8. | Ammeter ( Heater Ckt.) | 1 No. | 1 No. | 1 No. |
| 9. | Current Transformer | 3 No. | 3 No. | Nil |
| 10. | Potential Transformer | 1 Set | Nil | Nil |
| 11. | Energy meter | 1 No. | 1 No. | Nil |
| 12. | TTB | 1 No. | 1 No. | Nil |
| 13. | Indicating Lamp | 6 No. | 6 No. | 4 No. |
| 14. | MCB ( DC Ckt) | 1 No. | Nil | 1 No. |
| 15. | MCB ( AC Ckt, 16 Amps) | 1 No. | Nil | 1 No. |
| 16. | MCB ( Motor ckt.) | 1 No. | 1 No. | 1 No. |
| 17. | 3 Position PT Selector Switch (16 Amps) | 1 No. | 1 No. | Nil |
| 18. | Voltage Selector Switch | 1 No. | Nil | Nil |
| 19. | Electronic Buzzer (DC Volt operated) | Nil | Nil | 1 No. |
| 20. | Electronic Buzzer (AC Volt operated) | Nil | Nil | 1 No. |
| 21. | Heater with controller | 1 Set | 1 Set | 1 Set |
| 22. | T-N-C Switch | 1 No. | 1 No. | 1 No. |
| 23. | Local – Remote Switch | 1 No. | 1 No. | 1 No. |
| 24. | Mushroom type--Push Button for Emergency Trip | 1 No. | 1 No. | 1 No. |
| 25. | Push Button for Relay reset | 1 No. | 1 No. | Nil |
| 26. | Spring Charging Handle | 1 No. for every three Panel. | | |
| 27. | VCB Operating Handle | 1 No. for every three Panel | | |
| 28. | HRC Fuse & Link | As required | As required | As required |
| 29. | Trip Circuit supervision Relay | 1 No. | 1 No. | 1 No. |
| 30. | Push Button for Annunciator Accept & Reset | 2 Nos. | 2 Nos. | 2 Nos. |
| 31 | Push Button for Master trip relay reset. | 1 No. | 1 No. | 1 No. |
| 32 | DC Supervision Relay | Nil | Nil | 1 No. |

**NB: The Bidder has to include list of other Items for the Equipment to meet our Technical requirement.**

## Annexure-G

GURANTEED TECHNICAL PARTICULARS

( To be submitted by the Bidder )

| | | |
|---|---|---|
| 1. | **General :** | |
| | Name of the Company | |
| | Office address | |
| | Factory address | |
| 2. | **Panel** | |
| | Make | |
| | Type & Designation | |
| | Application Standard | |
| | Rated Voltage | |
| | Highest Voltage | |
| | Normal Current | |
| | Frequency | |
| | STC for 3 Sec. | |
| | Breaking Capacity | |
| | Making Capacity | |
| | Power frequency withstand voltage | |
| | Impulse withstand voltage | |
| | AC Aux. Voltage. | |
| | DC Aux. Voltage | |
| | Degree of Protection for HV & LV Compartment | |
| | Thickness of metal sheet ( mm ). | |
| | For load-bearing member | |

| | | |
|---|---|---|
| | For Doors & covers | |
| | Dimension of Incomer Panel in mm ( H x W x D ) | |
| | Dimension of Outgoing Panel in mm ( H x W x D ) | |
| | Dimension of Bus-Coupler & <br> Trunk Panel in mm ( H x W x D ) | |
| | **Single feeder panels to be supplied with Bus Bar for connection with adjacent Incomer / Bus Coupler / Feeder.** [ Yes / No ] | |
| | HV Cable termination height in the cable compartment | |
| **3.** | **Bus Bar** | |
| | Material | |
| | Shape | |
| | Cross sectional area of main Bus. | |
| | Cross sectional area of Riser. | |
| | Type of plating | |
| | Normal Current currying capacity | |
| | STC for 3 Sec. | |
| | Temp. Rise over ambient at normal current | |
| | Current density for main Bus & Riser | |
| | Phase to Phase clearance | |
| | Phase to ground clearance | |
| | Type of insulation | |
| | Colour Identification | |
| | Bus Barsleeve Full voltage sleeved with**shrouds** on joints | |
| **4.** | **Bus support insulator** | |

| | | |
|---|---|---|
| | Reference Standard. | |
| | Make | |
| | Material | |
| | Dry Power frequency Withstand Voltage for one minute | |
| | Wet Power frequency Withstand Voltage for one minute | |
| | Impulse Withstand voltage | |
| | Creepage distance | |
| **5.** | **Vacuum Circuit Breaker** | |
| | Make | |
| | Type | |
| | Reference Standard | |
| | Rated voltage | |
| | Highest voltage | |
| | Frequency | |
| | Normal Current | |
| | Breaking capacity | |
| | Making capacity | |
| | STC for 3 Sec. | |
| | Temp. Rise over ambient at normal current | |
| | No of break per phase. | |
| | Operating duty cycle | |
| | Single Phase Capacitor Breaking capacity | |
| | Three Phase Capacitor Breaking capacity | |
| | Line Charging Breaking capacity | |

| | | |
|---|---|---|
| | Cable Charging Breaking capacity | |
| | Closing time | |
| | Opening time | |
| | Mechanical Endurance capacity | |
| | Electrical Endurance capacity | |
| | Operating mechanism | |
| | Type of isolation & Drawout | |
| | Details of mechanical interlock provided | |
| | DC Wattage of Trip Coil. | |
| | DC Voltage range for Trip operation. | |
| | DC Wattage of Closing Coil. | |
| | DC Voltage range for Closing operation. | |
| | Details of electrical interlock. | |
| 6. | **Vacuum Bottle** | |
| | Make | |
| | Rated voltage | |
| | Type and model no. | |
| | Normal current | |
| | Breaking capacity | |
| | Making capacity | |
| | STC for 3 Sec. | |
| | Maximum contact separation length | |
| | Minimum Mechanical life in no. of operation | |
| | Minimum Electrical Life in no. of operation at rated normal current | |
| | Minimum Electrical Life in no. of operation at rated | |



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| | | |
|---|---|---|
| | full short circuit current | |
| | Power frequency withstand voltage (dry) | |
| | Impulse withstand voltage | |
| | Contact material | |
| | Type of plating | |
| | Contact pressure | |
| 7. | **Current Transformer** | |
| | Make | |
| | Reference Standard | |
| | Type | |
| | Voltage | |
| | Frequency | |
| | Insulation level | |
| | Class of insulation | |
| | Ratio (Incomer & Feeder) | |
| | Class of accuracy (Incomer & Feeder) | |
| | Burden (Incomer & Feeder) | |
| | STC for 1 Sec. | |
| | ISF of Metering Core at lower ratio | |
| | Vk & Magnetising Current(at lower ratio) | |
| | Fixing Arrangement | |
| 8. | **Potential Transformer** | |
| | Make | |
| | Reference Standard | |
| | Insulation level | |

| | | |
|---|---|---|
| | Class of insulation | |
| | Type | |
| | Frequency | |
| | Winding connection | |
| | Type of Core connection | |
| | Ratio | |
| | Class of accuracy | |
| | Burden per Phase | |
| | Over Voltage Factor for 8 hrs. | |
| | Fixing position. | |
| | Fixing arrangement | |
| | Primary Fuse rating | |
| **9.** | **O/C & E/F Relay** | |
| | Make | |
| | Type | |
| | Model no. | |
| | **Ordering Code (with supporting Data Sheet)** | |
| **10.** | **Trip Relay** | |
| | Make | |
| | Type | |
| | Model no. | |
| | **ORDERING CODE**<br>**(with supporting data sheet)** | |
| | **Whether Electrically Resettable**(Yes/No) | |
| **11.** | **Digital Ammeter ( CT operated )** | |
| | Make | |

| | | |
|---|---|---|
| | Type | |
| | Size | |
| | Basic CTR | |
| | Accuracy class | |
| | AC Aux. voltage | |
| | Fixing type. | |
| **12.** | **Digital Ammeter ( Heater circuit )** | |
| | Make | |
| | Type | |
| | Size | |
| | Current range | |
| | Accuracy class | |
| | AC Aux. voltage | |
| | Fixing type. | |
| **13.** | **Digital Volt meter ( PT operated )** | |
| | Make | |
| | Type | |
| | Size | |
| | PTR | |
| | Accuracy class | |
| | AC Aux. voltage | |
| | Fixing type. | |
| **14.** | **Analog Voltmeter ( DC Circuit )** | |
| | Make | |
| | Type | |



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| | | |
|---|---|---|
| | Size | |
| | Accuracy class | |
| | Fixing type. | |
| **15.** | **Energy meter** | |
| | Make | |
| | Type | |
| | Model no. | |
| **16.** | **Terminal connector** | |
| | Make | |
| | Type | |
| | Size | |
| **17.** | **Control Switch** | |
| | Make | |
| | Type | |
| | Contact configuration | |
| | AC & DC Current rating | |
| | AC Voltage rating. | |
| **18.** | **Spring Charge Limit Switch** | |
| | Make | |
| | Type | |
| | No of contact ( NO + NC ) | |
| | No of spare contact ( NO + NC ) | |
| | AC & DC Current rating | |
| | AC Voltage rating. | |
| **19.** | **Local Remote Switch.** | |

| | | |
|---|---|---|
| | Make | |
| | Type | |
| | Contact configuration | |
| | AC & DC Current rating | |
| | AC Voltage rating. | |
| **20.** | **Spring charging motor** | |
| | Make | |
| | Type | |
| | Rated AC Voltage | |
| | Rated wattage | |
| | AC Voltage variation for operation | |
| **21.** | **Operation counter** | |
| | Make | |
| | Type | |
| | Operation count range | |
| **22.** | **Space heater** | |
| | Make | |
| | Type | |
| | Rated AC Voltage | |
| | Wattage | |
| **23.** | **Thermostat** | |
| | Make | |
| | Setting range | |
| **24.** | **Indication Lamp** | |
| | Make | |

| | | |
|---|---|---|
| | Type | |
| 25. | **Auxiliary contact switch** | |
| | Make | |
| | Type | |
| | No of contact ( NO + NC ) | |
| | No of spare contact ( NO + NC ) | |
| | AC & DC Current rating | |
| | AC Voltage rating. | |
| 26. | **Annunciator** | |
| | Make | |
| | No of window. | |
| | Aux voltage | |
| | No of inbuilt push switch. | |
| 27 | **Voltage selector switch** | |
| | Make | |
| | Type | |
| 28 | PT **selector switch** | |
| | Make | |
| | Type | |
| | Current Rating | |
| 29 | **TTB** | |
| | Make | |
| | Type | |
| | Size | |
| 30 | **Terminal Block** | |

| | | |
|---|---|---|
| | Make | |
| | Type | |
| 31 | **MCB** | |
| | Make | |
| | Type | |
| 32 | **Fuse & Link Unit** | |
| | Make | |
| | Type | |
| 33 | **HRC Fuse** | |
| | Make | |
| | Type | |
| 34. | **Control wire** | |
| | Make | |
| | Type | |
| | Voltage Grade | |
| | Size ( Sq. mm) | |
| | For CT & PT | |
| | For control indication & annunciation circuit | |
| | For main AC & DC and Bus Wiring | |
| | Colour | |
| 35. | **Earth Bus** | |
| | Material | |
| | Shape | |
| | Size | |
| 36. | **Painting details** | |

| | | |
|---|---|---|
| | Paint type | |
| | Paint shade | |
| | Paint thickness | |
| **37.** | **Shipping dimension of equipment (HxWxD)** | |
| **38.** | **Total weight of the equipment** | |
| **39.** | **Accessories** | |
| | Spring Charging Handle | |
| | VCB Operating Handle | |
| **40.** | Guarantee of the total equipment including any integral part of the equipment | |

| | | |
|---|---|---|
| 1 | State clearly about compliance of following provisions as per provision in relevant Clauses of Specification. | |
| a) | Name, Rating, Making, Property Plate, Circuit Level, Connection Diagram | |
| b) | Operation Instruction Chart | |
| 2 | State clearly information about type testing of similar Equipment in recognized Institution as per Technical Specification. If yes, mention clearly nature of tests carried out with Name of Testing Institution. | |
| 3 | Information about testing facilities available at the Manufacturer's Works for conducting Routing/Acceptance Tests. | |

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| 4 | Information about submission of following documents with the Tender | |
| a) | Copies of Type Test Reports. | |
| b) | Copies of Tender Drawing Literature & Manual. | |
| c) | Copies of Tender Performance Reports & Credential | |
| d) | Copies of Tender Literature & Leaflets of Device | |
| e) | List of Recommended Spares. | |
| 5 | Information about submission of Deviation Sheet | |



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## ANNEXURE-H

### GTP for Numerical Feeder (Non directional) Protection Relay

| Sl. No. | Feature and Function | Bidders response |
|---|---|---|
| 1.1 | **Make** | |
| | Type | |
| | **ORDERING CODE**<br>**(with supporting data sheet)** | |
| 1.2 | Conformance to<br>i. IEC255-4 | |
| | ii. IEC 61850 | |
| 1.3 | No. of CT inputs for O/C and E/F Protection | |
| 1.4 | Type test report submitted(y/n) | |
| 1.5 | Relay shall be of Numeric Design | |
| 1.6 | Relay designed for bay protection and Control | |
| 1.7 | Size of Relay LCD screen | |
| 1.8 | Relay is equipped with CB close and open key/push buttons | |
| 1.9 | Relay has following protection functions:<br>a. Three phase over current<br>b. Earth fault<br>c. Thermal overload function<br>d. Broken conductor protection function<br>e. Circuit Breaker Maintenance function | |
| 2. | a. One time delayed element and two high set elements<br>b. Setting range and step for IDMT element for both current and Time Multiplier Setting | |
| | c. Selectable Current/Time Curve for IDMT element | |
| | d. Setting range and step for high set elements for both current and time delay | |
| 3. | Sampling rate and frequency of analog signal | |
| 4. | Whether remote controllable from SCADA | |
| 5. | a. No. of Digital Inputs<br>b. Voltage rating of Digital Inputs<br>c. Provision of testing without current injection | |

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|---|---|---|
| 6. | Supervision for CB open and Closed status | |
| 7. | No. of programmable LEDs and no. of Latched LEDs | |
| 8. | Analog Measurement and display supported | |
| 9. | Fault Record storage capacity | |
| 10. | Event storage capacity | |
| 11. | Disturbance record storage capacity | |
| 12. | MMI with keypad and LCD provided | |
| 13. | Rated DC Supply and tolerance | |
| 14. | Rating of CT/PT secondary | |
| 15. | Rated frequency | |
| 16. | a. Operating ambient temperature & humidity | |
| | b. Withstanding capability of Electromagnetic interference as per relevant part of IEC 61850 | |
| 17. | Mounting | |
| 18. | Watchdog | |
| 19. | a. Nominal Feeder current | |
| | b. CT Ratio setting | |
| | c. Earth fault current with time delay IEC Curves, 2nd stage for instantaneous trip(less than 50 ms) | |
| | d. High set with delay | |
| | e. IEC Curves for all O/C and E/F have user selectable? | |
| 20. | a. No. of Digital Output Contacts | |
| | b. Contact rating | |
| 21. | Mode of Time Synchronization | |
| 22. | Type of Lugs and terminators | |
| 23. | MTBF | |
| 24. | Lifespan | |
| 25. | Compliance to Type Test | |
| 26. | Communication Port<br>a. Rear port- details<br>b. Front port-details | |
| 27. | Whether Communication Ports are native to the Relay | |
| 28. | Protocol supported for Rear Port | |

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|---|---|---|
| 29. | Protocol supported for Front port | |
| 30. | Start and trip output contacts are freely programmable | |
| 31. | Cable for connection of Relay to laptop(USB port) along with converter and power supply if required for relay local setting | |
| 32. | Basic application software for setting change, parameterisation | |
| 33. | CD with software(licensed ) to download disturbance recorder, event log and evaluation of those records | |
| 34. | Graphical configuration tool for I/P, O/P and functional building block for protection and control | |
| 35. | Any other software required for integration with SCADA. | |

**ANNEXURE-J**

## GTP for Master Trip Relay

| Sl. No. | Description | Bidder's Response |
|---|---|---|
| 01. | **Manufacturer Name** | |
| 02. | **Type and designation** | |
| 03. | **ORDERING CODE (with supporting data sheet)** | |
| 04. | **Electrical reset** | |
| 05. | **Mounting** | |
| 06. | **High Burden relay** | |
| 07. | **Operating Time** | |
| 08. | **Rated DC supply and tolerance** | |
| 09. | **No. of NO Contact** | |
| 10. | **No. of NC Contact** | |

**Date :**

**Signature:**
**Name:**
**Designation:**
**Company Seal:**



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## ANNEXURE-E

**All numerical relays shall be cyber security compliant; in line with the guidelines of the Ministry of power, Govt. of India.**

### Technical Specification for Non-directional over current and earth fault relay

| Sl. No. | Feature and Function | Technical requirement |
|---|---|---|
| 1 | Purpose and application | It is intended to automate the Switchgears specified in the scope of supply and use Communicable Numeric relays for Protection, Control, Metering and Status monitoring. This specification is based on the understanding that an integrated Automation System along with protections shall be provided and same shall have provisions for Integration with SCADA system. All the feeders shall be remote controlled from Employer's SCADA and from the local console of the numerical relays.<br><br>Numerical multifunctional combined Microprocessor based Feeder protection and management relay to protect the 11 kV Feeder from all electrical and other faults along with reporting system, Disturbance record for fault analysis. Bidder should comply with any especial requirement or feature asked for retrofitting the relays. Relay should be IEC 61850 compliant. Relay should have 4 CT input for O/C and E/F protection. There should be option for derivation of E/F internally. |
| 2. | Main Protection Feature | 1. Relay should have minimum two group of setting. Setting group changeover required from digital status input.<br>2. Electrical over load protection with selectable IEC curves with two stage, first stage to be used as Definite Time / IDMT and second stage to be used as high set for short circuit protection.<br>3. Earth fault protection in two stages with IEC characteristics. First stage to be used as IDMT/Definite Time and second stage to be used as instantaneous elements. Earth fault element should be suitable for both CBCT and residual type CT connection.<br>4. Negative phase sequence Protection with IEC Curves.<br>5. CB Fail Protection & time settable as per user.<br>6. The relay should be immune to DC switching while carrying current i.e. no spurious trip should be generated if relay DC is made On and Off<br>7. The relay should conform to the IEC255-4 or BS 142 for Inverse time characteristics.<br>8. The relay should have features to monitor for broken conductor and CB opening time. |
| 3. | Processor feature | Relay shall be completely Numerical with protective elements having software algorithm based on sampling of analog inputs. Sampling Rate of Analog Signal: The sampling rate should be 1000 Hz for 50 Hz signal or better for each analog channel. Hardware based |

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|---|---|---|
| | | measurements shall not be acceptable. |
| 4. | Operational Philosophy | The operation of Relay shall be possible both locally from the Switchgear and remote & Local Work station. The local position shall be displayed in remote / local workstation and remote operation shall be blocked if the switch is in Local. Clear control priorities shall prevent initiation of operation of a single switch at the same time from more than one of the various control levels and there shall be interlocks among various control levels. The priority shall always be with the lowest enabled control level. Relay accuracy shall not be affected by system frequency fluctuation. |
| 5. | Status/Optical Inputs/Digital inputs | 1. Minimum 10number status inputs are required<br>2. All status inputs should be 30 V DC.<br>3. Setting group is required to be changed with any Digital input status.<br>4. Trip circuit supervision with DI status<br>5. The digital inputs shall be acquired by exception with 1ms resolution. Contact bouncing in digital inputs shall not be assumed as change of state.<br>6. Relay should have comprehensive self diagnostic feature with remote indication of relay failure and alarm shall be generated without tripping of circuit<br>7. Provision of Testing output relays without any current injection.<br>8. No. of programmable LEDs- at least 8nos. with latching option. |
| 6. | Main measuring and reporting feature | All measurements should be in primary quantities. Minimum following displays are required in alpha numeric:-<br>1. Three phase (Positive sequence) current<br>2. Neutral(zero sequence) current<br>3. All the trips should have clear indication on the relay terminals<br>4. Resetting should be selectable as hand reset or auto reset.<br>5. The default relay LCD shall be user defined to display primary circuit loading. |
| 7. | Memory and Recording Feature | 1. The relay setting and programming should be stored in EEPROM so that during Aux. Power failure the said data is not lost.<br>2. Relay should have event log, trip log and DR record. All logs should go in to history.<br>3. All tripping of relay should initiate DR in auto without extra binary input. Triggering of DR with binary input should be user configurable.<br>4. The last 2 fault DR records should be in flash memory and DR will not erase in case of DC supply fail for more than 2 days.<br>5. Should be able to record at least 5 Oscillographic disturbances and 5 fault records and 250 event records.<br>6. Minimum Four no. of latest trip log with cause of trip should be stored in memory along with date and time stamping. The memory should not be lost with the switching off of DC.<br>7. The relay should have fault-recording feature with current waveform and Digital Input status. The fault waveform should consist of minimum four current waveforms of three phases current and zero sequence current and DI status. Triggering time for Pre and Post should have user |

| | | |
|---|---|---|
| | | selectable. This record should be in flash memory for minimum 7(seven) days even after switching off the DC supply.<br>8. The fault should be date and time stamped.<br>9. Communication protocol IEC 61850. |
| 8. | Auxiliary Supply | 30 V DC to - 25% to + 10%, 2 wire unearthed system. Necessary software shall be in-built for proper shutdown and restart in case of power failure. Auxiliary supply burden will be around 20Watt. |
| 9. | Rated CT/PT secondary | 5/1 Amp(User selectable) , CTs used to be protection class |
| 10. | Rated frequency | 50 HZ +/- 3% |
| 11. | Ambient condition | 1. Operating ambient temperature upto 55 Deg C<br>2. Operating Humidity upto 100 %<br>3. Relay shall meet the requirement for withstanding electromagnetic interference according to relevant parts of IEC 61850. Failure of single component within the equipment shall neither cause unwanted operation nor lead to a complete system breakdown. |
| 12. | Module and Mounting | 1. Relay should be flush mounted type<br>2. If module is draw-out type then it should have CT shorting facility of make before break type.<br>3. Mounting in switchgears located in non AC rooms.<br>4. Galvanic isolation between field connection and relay hardware should be there. |
| 13. | Watchdog and self monitoring | The relay should have facility to monitor the healthiness of its circuits and components by own monitoring system. In case of any problems, the alarm should be generated by one of the output contacts. The alarm as soft signal to be sent to SCADA system as well. Necessary support documentation explaining the self diagnostic feature shall be furnished. |
| 14. | Relay parameterisation and Settings | Settings possible through relay keypad relay HMI software should be as follows:-<br>1. Nominal Feeder current 2% to 110 %<br>2. CT Ratio setting 10-1000(approx.)<br>3. Earth fault current 5 to 40% with time delay IEC Curves, 2nd stage for instantaneous trip(less than 50 ms)<br>4. Over current trip- 50% to 200% of 1/5 Amp with time delay as per IEC Curves.<br>5. High set with delay 200% to 2000%<br>6. IEC Curves for all O/C and E/F have user selectable. |
| 15. | Output Relays | Minimum 10number output relays are required out of which<br>1. One potential free change over contact should be provided for start inhibit of relay.<br>2. All o/p contact should be freely programmable.<br>3. Rating of trip contacts:-<br>a) Contact durability>10K operation<br>b) 15 Amp make and carFry for 3 sec for trip contact<br>c) Make and carry for trip contacts L/R<=40ms<br>4. Rating of Alarm contacts:- 5 Amp make and carry continuously for 5 sec.<br>Testing of Output relays through keypad on relay fascia and relay HMI software. Output relay dwell time shall be user programmable or fixed at 100ms. |

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|---|---|---|
| 16. | Relay software and Man Machine Interface | 1. The relay should have native IEC 61850 Communication Protocol.<br>2. Should have password protected key padlock.<br>3. Necessary software for relay setting , retrieving DR, event log, trip log should be supplied by the Bidder. Necessary License is to be issued for WBSEDCL, if required.<br>4. Bidder has to supply communication hardware for relay setting, DR downloading from front port. This device should be compatible to USB/Ethernet port.<br>5. It shall be possible to transfer the data stored in the DFR to computer on IEEE/COMTRADE format. The data format shall be compatible for dynamic protection relay testing on relay test kit. COMTRADE data viewing software to be provided.<br>6. Multiuser/Corporate license for installation on minimum 7 nos. of PCs. |
| 17. | Date and time | Date and Time stamping with faults and record. The clock should be powered from internal cell and should not required setting after every DC switching. The internal cell life minimum 5 years. Time synchronization by IRIG-B or SNTP. For time synchronization through SNTP is to be provided from clock signal coming from RTU. In case of IRIG-B, time synchronization will be done with GPS clock signal from GPS receiver located at substation. |
| 18. | Lugs and terminators | All CT and PT terminals shall be provided as fixed (screwed) type terminals on the relay to avoid anyhazard due to loose connection leading to CT opening or any other loose connection. All CT terminals of the Relay to be compatible with ring type lug. Necessary amount of lugs should be supplied along with each relay for CT connection and control wiring. |
| 19. | Manuals, Drawings and Literature | 1. The relays should be supplied with manuals with all technical and operating instructions.<br>2. All the internal drawings indicating the logics and block diagram details explaining principle of operation should be given at the time of supply.<br>3. Mapping details shall be submitted in IEC format. |
| 20. | Assistance in commissioning | As part of this offer, Bidder shall have to provide necessary support during integration of relays with the third party SCADA system which will be commissioned in future. |
| 21. | Standard documentation per Relay, according to IEC 61850 | 1. MICS document(model implementation conformance statement)<br>2. PICS(protocol implementation conformance statement)<br>3. PIXIT document<br>4. Conformance Test certificate by KEMA/CPRI.<br>All the above mentioned certificates shall be submitted along with Techno-commercial Bid.<br>5. ICD file<br>6. SCD file |
| 22. | Extendibility in Future | The Bidder shall provide all necessary software tools along with source codes to perform addition of bays in future and complete integration with SCADA by the User. These software tools shall be able to configure relay, add analog variable, alarm list, event list, modify interlocking logics etc. for additional |

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| | | bays/equipments which shall be added in future. |
| 23. | Lifespan | The supplier should mention following:-<br><br>1. Product maturity: The Bidder should mention the time period for which the product is in the market<br>2. Expected production life<br>3. Hardware/Firmware change notification process. Upgrades to be provided free of cost within the warranty period/5 years whichever is later, if needed.<br>4. Lifespan of standard tools and processes for relay configuration, querying and integration. |
| 24. | Standards | The relay should conform to the IEC255-5 or equivalent BS / ANSI for following:-<br><br>1. Overload withstand test<br>2. Dielectric withstand: 2kV in common, 1 kV in differential mode<br>3. Impulse Voltage: 5kV in common, 1kV in differential mode<br>4. Insulation resistance>100 M ohm.<br>5. Vibration: Shock and bump and Seismic<br>6. Storing and transportation<br>7. Radio Interference: IEC 61000 for high frequency disturbance, Transient disturbance, Electrostatic discharge<br>8. KEMA/CPRI Certification for the particular model offered with respect to IEC61850 Protocol. |
| 25. | Communication Port | 6. **Two nos. IEC 61850 (Edition-2) protocol compliant 100 MBPS Ethernet RJ45/F.O ports having selectable HSR (High Availability Seamless Redundancy) and PRP (Parallel Redundancy Protocol) Protocol compliance for communication with SCADA system through two managed Ethernet Switches operating in redundant mode.** The communication shall be made in 1+1 mode between individual IED to Switch, such that failure of one set of LAN shall not affect the normal operation of SCADA. However, it shall be alarmed in SCADA.<br>The relay rear ports should accommodate both daisy chain and star bus topology for SCADA integration.<br><br>1. Functioning of Relay shall not hamper to fault occurring any interconnected relay.<br>2. One Front port Ethernet RJ45/USB 2.0 for relay parameterization and configuration etc. with the help of PC.<br>3. Relay should generate GOOSE message as per IEC 61850 standard for interlocking and also ensure interoperability with third party relays. |
| 26. | Name Plate and marking | Each IED shall be clearly marked with manufacturer's Name, type, serial no. and electrical rating data. Name plates shall be made of anodized aluminium with white engraving on black surface. |

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| 27. | Performance Guarantee | Relays will be guaranteed for the period of five years from the date of last<br>dispach.<br>Any problem in the said period should be attended free of charge inclusive of repair/replacement of relays/ component (both H/W, S/W). |
| 28. | Type Test | • Dielectric Withstand Test—IEC 60255-5<br>• High Voltage Impulse Test, class III --- IEC 60255-5(5kV peak, 1.2/50 micro Sec;3 Positive and 3 negative shots at interval of 5 Sec.)<br>• DC Supply Interruption ---- IEC 60255-11<br>• AC Ripple on DC supply ---- IEC 60255-11<br>• Voltage Dips and Short Interruptions --- IEC 61000-4-11<br>• High frequency Disturbance ---- IEC 60255-22-1, Class III<br>• Fast Transient Disturbance ---- IEC 60255-22-4, Class-IV<br>• Surge withstand capability ---- IEEE/ANSI C 37.90.1(1989)<br>• Degree of Protection<br>• Electromagnetic compatibility<br>• Mechanical stress/vibration test<br>• Temperature withstand<br>Type test reports for the above tests shall be submitted for the approval of WBSEDCL along with Tender. Wherever the above mentioned standards and IEC 61850 overlap, the latter will prevail. |
| 29. | Credential as pre-requisite of Tender | Qualifying Requirement:-<br>1 . Successfully tested and in operation for minimum one year in WBSEB /WBSEDCL / WBSETCL network.<br>OR<br>Copies of performance certificate for minimum two years successful operation of the offered relay in respect to implementation of IEC 61850 Protocol to any SCADA / Sub Station Automation System form different Organisations specially in any Power Utility in India as on the due date of bid opening.<br>2 . Documentary evidence for being manufacturers like registration Certificate issued by SSI / NSIC / Directorate of Industries / DGS&D, etc. for qualifying requirement.<br>3 . The bidder should have testing facilities of all functional tests or should have arrangement of all functional tests at Government approved Testing Laboratories.<br>4 . Successful Inter Operability and functional Test :-<br>Inter Operability Test of offered relay, which complies with other technical aspects of WBSEDCL'S Technical Specifications, with the existing relays in WBSEDCL Network will be tested in Distribution Testing Department, WBSEDCL for which due intimation for supply of sample of offered relay will be given to the bidder after opening of the Techno-Commercial bid. The bidder needs to submit the said relay to Distribution Testing Department, Abhikshan within 15 (fifteen ) days from the said date of intimation. |
| 30. | Offer for relay | **Offer with all relevant documents for numerical relay may be submitted with different make as per Annexure-B alongwith the Bidbut after placement of order, drawing for approval needs to be submitted with only one type of accepted offered relay.** |

## Technical specification for IEC 61850 compliant Directional O/C and E/F Relay

| Sl. No. | Feature and Function | Technical requirement |
|---|---|---|
| 1. | Purpose and application | It is intended to automate the Switchgears specified in the scope of supply and use Communicable Numeric relays for Protection, Control, Metering and Status monitoring. This specification is based on the understanding that an integrated Automation System along with protections shall be provided and same shall have provisions for Integration with SCADA system. All the feeders shall be remote controlled from WBSEDCL's SCADA and from the local console of the numerical relays.<br><br>Numerical multifunctional combined Microprocessor based Feeder protection and management relay to protect the 33 kV Feeder from all electrical and other faults along with reporting system, Disturbance record for fault analysis. Bidder should comply with any especial requirement or feature asked for retrofitting the relays. Relay should be IEC 61850 compliant. Relay should have 4 CT input, 3 input for O/C and residual E/F protection will be derived internally. One CT input may be used for unbalanced current protection. Relay should have 4 voltage input, 3 input for VT element for directional O/C protection with internally derived residual voltage for E/F protection. Another VT input will be used for residual voltage protection. Relay should have two stage over voltage and under voltage protection. **The Directional O/C & E/F Relay should have the Site-selectable provision for operating in both Directional & Non-directional mode depending on site requirement.** |
| 2. | Main Protection Feature for directional O/C & E/F relay. | 1. Electrical over load protection with selectable IEC curves with two stage, first stage to be used as Definite Time / IDMT and second stage to be used as high set for short circuit protection.<br>2. Earth fault protection in two stages with IEC characteristics. First stage to be used as IDMT/Definite Time and second stage to be used as instantaneous elements. Earth fault element should be suitable for both CBCT and residual type CT connection.<br>3. Negative phase sequence Protection with IEC Curves.<br>4. CB Fail Protection & time settable as per user.<br>5. The relay should be immune to DC switching while carrying current i.e. no spurious trip should be generated if relay DC is made On and Off<br>6. The relay should conform to the IEC255-4 or BS 142 for Inverse time characteristics.<br>7. VT fuse fail detection on NPS current/NPS Voltage or zero sequence current/voltage based logic and blocking of under voltage protection by VT fuse fail detection. |

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| | | 8. Three phase VT fuses fail detection on current based logic.<br>9. The relay should have features to monitor for broken conductor and CB opening time.<br>10. The relay shall be designed for application in WBSEDCL's distribution network where the system is non-effecively earthed through earthing transformer emanating at 33kV bus of 132/33 kV substation.<br>11. Relay should have minimum two group of setting. Setting group changeover required from digital status input. |
| 3. | Processor feature | Relay shall be completely Numerical with protective elements having software algorithm based on sampling of analog inputs. Sampling Rate of Analog Signal: The sampling rate should be 1000 Hz for 50 Hz signal or better for each analog channel. Hardware based measurements shall not be acceptable. |
| 4. | Operational Philosophy | The operation of Relay shall be possible from both locally from the Switchgear and remote and Local Work station. The local position shall be displayed in remote / local workstation and remote operation shall be blocked if the switch is in Local. Clear control priorities shall prevent initiation of operation of a single switch at the same time from more than one of the various control levels and there shall be interlocks among various control levels. The priority shall always be with the lowest enabled control level. Relay accuracy shall not be affected by system frequency fluctuation. |
| 5. | Status/Optical Inputs/Digital inputs | 1. Minimum 10number status inputs are required<br>2. All status inputs should be 30 V DC<br>3. Setting group is required to be changed with any Digital input status.<br>4. Trip circuit supervision with DI status<br>5. The digital inputs shall be acquired by exception with 1ms resolution. Contact bouncing in digital inputs shall not be assumed as change of state.<br>6. Relay should have comprehensive self diagnostic feature with remote indication of relay failure and alarm shall be generated without tripping of circuit<br>7. Provision of Testing output relays without any current injection.<br>8. No. of programmable LEDs- at least 8nos. with latching option. |
| 6. | Main measuring and reporting feature | All measurements should be in primary quantities. Minimum following displays are required in alpha numeric:-<br><br>1. Three phase (Positive sequence) current, Three phase voltage<br>2. Neutral (zero sequence) current, MW, MVAR, Frequency, Pf, MVA etc.<br>3. All the trips should have clear indication on the relay |

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| | | terminals<br>4. Resetting should be selectable as hand reset or auto reset. |
| 7. | Memory and Recording Feature | 1. The relay setting and programming should be stored in EEPROM so that during Aux. Power failure the said data is not lost.<br>2. Relay should have event log, trip log and DR record. All logs should go in to history.<br>3. All tripping of relay should initiate DR in auto without extra binary input. Triggering of DR with binary input should be user configurable.<br>4. The last 2 fault DR records should be in flash memory and DR will not erase in case of DC supply fail for more than 2 days.<br>5. Should be able to record at least 5 Oscillographic disturbances and 5 fault records and 250 event records.<br>6. Minimum Four no. of latest trip log with cause of trip should be stored in memory along with date and time stamping. The memory should not be lost with the switching off of DC.<br>7. The relay should have fault-recording feature with current waveform and Digital Input status. The fault waveform should consist of minimum four current waveforms of three phase current and zero sequence current and DI status. Triggering time for Pre and Post should have user selectable. This record should be in flash memory for minimum 7 days even after switching off the DC supply.<br>8. The fault should be date and time stamped.<br>9. Communication protocol IEC 61850. |
| 8. | Auxiliary Supply | 30 V to - 25% to + 10%, 2 wire unearthed system. Necessary software shall be in-built for proper shutdown and restart in case of power failure. Auxiliary supply burden will be around 20Watt. |
| 9. | Rated CT/PT secondary | 5/1 Amp(site selectable) , CTs used to be protection class.<br><br>3PT input rated 110 Volt (L-L) |
| 10. | Rated frequency | 50 HZ +/- 5% |
| 11. | Ambient condition | 1. Operating ambient temperature up to 55 Deg C<br>2. Operating Humidity up to 100 %<br>3. Relay shall meet the requirement for withstanding electromagnetic interference according to relevant parts of IEC 61850. Failure of single component within the equipment shall neither cause unwanted operation nor lead to a complete system breakdown. |
| 12. | Module and Mounting | 1. Relay should be flush mounted type<br>2. If module is draw out type then it should have CT shorting facility of make before break type.<br>3. Mounting in switchgears located in non AC rooms.<br>4. Galvanic isolation between field connection and relay hardware should be there. |

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| 13. | Watchdog and self monitoring | The relay should have facility to monitor the healthiness of its circuits and components by own monitoring system. In case of any problems, the alarm should be generated by one of the output contacts. The alarm as soft signal to be sent to SCADA system as well. Necessary support documentation explaining the self diagnostic feature shall be furnished. Watch dog contact shall be provided in addition to required 7BI and 7 BO. |
| 14. | Settings | Approximate settings possible should be as follows:-<br>1. Nominal Feeder current 2% to 110 %<br>2. CT Ratio setting 10-1000(approx.)<br>3. Earth fault current 5 to 40% with time delay IEC Curves, 2nd stage for instantaneous trip(less than 50 ms)<br>4. Over current trip- 50% to 200% of 1/5 Amp with time delays as per IEC Curves.<br>5. High set with delay 200% to 2000%<br>6. IEC Curves for all O/C and E/F have user selectable.<br>7. Selectable MTA for Directional features for O/C relay should cover 1st quadrant for effectively grounded system/ impedance grounded system/solid grounded system. |
| 15. | Output Relays | Minimum 10 number output relays are required out of which<br>1. One potential free change over contact should be provided for start inhibit of relay.<br>2. All o/p contact should be freely programmable.<br>3. Rating of trip contacts:-<br>a) Contact durability>10K operation<br>b) 15 Amp make and carry for 3 sec for trip contact<br>c) Make and carry for trip contacts L/R<=40ms<br>Rating of Alarm contacts:-<br>d) 8 Amp make and carry continuously for 5 sec.<br>Testing of Output relays through keypad on relay fascia and relay HMI software. Output relay dwell time shall be user programmable or fixed at 100ms. |
| **16.** | **Relay software and Man Machine Interface** | 1. The relay should have native IEC 61850 Communication Protocol.<br>2. Should have password protected key padlock.<br>3. Necessary software for relay setting, retrieving DR, event log, trip log, and downloading waveform should be supplied by the Bidder. Necessary Licensed is to be issued for |

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| | | WBSEDCL, if required.<br>4. Bidder has to supply communication hardware for relay setting, DR downloading from front port. This device should be compatible to USB/Ethernet port.<br>5. It shall be possible to transfer the data stored in the DFR to computer on IEEE/COMTRADE format. The data format shall be compatible for dynamic protection relay testing on relay test kit. COMTRADE Data viewer software is to be provided.<br>6. Multiuser/Corporate license for installation on minimum 7 nos. of PCs. |
| **17.** | **Date and time** | Date and Time stamping with faults and record. The clock should be powered from internal cell and should not required setting after every DC switching. The internal cell life minimum 5 years. Time synchronization by IRIG-B or SNTP. For time synchronization through SNTP is to be provided from clock signal coming from RTU. In case of IRIG-B, time synchronization will be done with GPS clock signal from GPS receiver located at substation. |
| **18.** | **Lugs and terminators** | All CT and PT terminals shall be provided as fixed (screwed) type terminals on the relay to avoid any hazard due to loose connection leading to CT opening or any other loose connection. Necessary amount of lugs should be supplied along with each relay for CT connection and control wiring.<br><br>All CT terminals should be capable to connect with ring socket. |
| **19.** | **Manuals, Drawings and Literature** | 1. The relays should be supplied with manuals with all technical and operating instructions.<br>2. All the internal drawings indicating the logics and block diagram details explaining principle of operation should be given at the time of supply.<br>3. Mapping details shall be submitted in IEC format. |
| **20.** | **Standard documentation per Relay, according to IEC 61850** | a.i.1. ICS document (model implementation conformance statement)<br>a.i.2. PICS(protocol implementation conformance statement<br>a.i.3. Conformance Test certificate from KEMA/CPRI.<br>a.i.4. PIXIT document<br>All the above mentioned certificates shall be submitted along with Techno-commercial Bid.<br>a.i.5. ICD file<br>a.i.6. SCD file |
| **21.** | **Extendibility in Future** | The Bidder shall provide all necessary software tools along with source codes to perform addition of bays in future and complete integration with SCADA by the User. These software |



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| | | tools shall be able to configure relay, add analog variable, alarm list, event list, modify interlocking logics etc. for additional bays/ equipments which shall be added in future. |
| **22.** | **Lifespan** | The supplier should mention following:-<br><br>1. Product maturity: The Bidder should mention the time period for which the product is in the market<br>2. Expected production life<br>3. Hardware/Firmware change notification process. Upgrades to be provided free of cost within the Guarantee period/5 years whichever is later, if needed.<br>4. Lifespan of standard tools and processes for relay configuration, querying and integration. |
| **23.** | **Standards** | The relay should conform to the IEC255-5 or equivalent BS / ANSI for following:-<br><br>1. Overload withstand test<br>2. Dielectric withstand: 2kV in common, 1 kV in differential mode<br>3. Impulse Voltage: 5kV in common, 1kV in differential mode<br>4. Insulation resistance>100 M ohm<br>5. Vibration: Shock and bump and Seismic<br>6. Storing and transportation<br>7. Radio Interference: IEC 61000 for high frequency disturbance, Transient disturbance, Electrostatic discharge<br>8. KEMA/CPRI Certification for the particular model offered with respect to IEC61850 Protocol |
| **24.** | **Communication Port** | 1. Two nos. IEC 61850 (Edition-2) protocol compliant 100 MBPS Ethernet RJ45/F.O ports having selectable HSR (High Availability Seamless Redundancy) and PRP (Parallel Redundancy Protocol) Protocol compliance for communication with SCADA system through two managed Ethernet Switches operating in redundant mode. The communication shall be made in 1+1 mode between individual IED to Switch, such that failure of one set of LAN shall not affect the normal operation of SCADA. However, it shall be alarmed in SCADA.<br>The relay rear ports should accommodate both daisy chain and star bus tropology for SCADA integration.<br><br>2. Functioning of Relay shall not hamper to fault occurring any interconnected relay.<br>3. One Front port Ethernet RJ45/USB 2.0 for relay parameterization and configuration etc. with the help of PC. In case RS-232 port offered, suitable interfacing cable |

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| | | with one end having RS 232 port and other end USB 2.0 to be provided to connect with PC free of cost.<br>4. Relay should generate GOOSE message as per IEC 61850 standard for interlocking and also ensure interoperability with third party relays. |
| 25. | **Name Plate and marking** | Each IED shall be clearly marked with manufacturer's Name, type, serial no. and electrical rating data. Name plates shall be made of anodized aluminium with white engraving on black surface. |
| 26. | **Performance Guarantee** | Relays will be guaranteed for the period of five years from the date of last dispach.<br><br>Any problem in the said period should be attended free of charge inclusive of repair/replacement of relays/ component (both H/W, S/W). |
| 27. | **Type Test** | • Dielectric Withstand Test—IEC 60255-5<br>• High Voltage Impulse Test, class III --- IEC 60255-5(5kV peak, 1.2/50 micro Sec;3 Positive and 3 negative shots at interval of 5 Sec.)<br>• DC Supply Interruption ---- IEC 60255-11<br>• AC Ripple on DC supply ---- IEC 60255-11<br>• Voltage Dips and Short Interruptions --- IEC 61000-4-11<br>• High frequency Disturbance ---- IEC 60255-22-1, Class III<br>• Fast Transient Disturbance ---- IEC 60255-22-4, Class-IV<br>• Surge withstand capability ---- IEEE/ANSI C 37.90.1(1989)<br>• Degree of Protection<br>• Electromagnetic compatibility<br>• Mechanical stress/vibration test<br>• Temperature withstand<br>Type test reports for the above tests shall be submitted for the approval of WBSEDCL along with Tender. Wherever the above mentioned standards and IEC 61850 overlap, the latter will prevail. |
| 28. | Credential as pre-requisite of Tender | 1. Copies of performance certificate for two years successful operation as on the due date of bid opening for the offered relay in respect to implementation of IEC 61850 protocol to any SCADA/substation automation system from reputed Power Sector Utility in India shall have to be furnished along with the Bid. Copies of Purchase Orders and corresponding Delivery Challans /Stores Receipt vouchers/ Excise Duty Invoice, etc., i.e. Proof of Execution of the Purchase Orders.<br><br>OR<br><br>Successful testing and operation of minimum one year in |

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| | | WBSEB/WBSEDCL/WBSETCL network..<br><br>2. Documentary evidence for being manufacturers like registration Certificate issued by SSI/NSIC/Directorate of Industries/DGS&D, etc. for Qualifying requirement.<br><br>3. The bidder should have testing facilities of all functional tests or should have arrangement of all functional tests at government approved testing laboratories.<br><br>**Inter-operability test:-**<br><br>After fulfilment of the above Q.R. inter-operability test of the offered relay (other than Make & Model used in WBSEDCL) with the existing relay in WBSEDCL Network will be tested in WBSEDCL Distribution Testing Department, WBSEDCL for which due intimation for supply of sampled of offered relay will be given to the Bidder. The Bidder needs to submit the said relay to Distribution Testing Department, WBSEDCL within one week from the said intimation.<br><br>The offered relay will only be accepted after fulfilment of above Q.R. &successful inter-operability test at WBSEDCL system. |

**Checklist for Bill of Material for supply**

| Sl. No. | Material | |
|---|---|---|
| 1. | Relay (Model No.) | Qty as per Tender |
| 2. | Lugs suitable for current and control, wiring | Qty as per Tender X Number of TBs in relay + 20% extra. |
| 3. | Cable for connection of Relay to laptop(USB port). Along with converter and power supply if required for relay local setting | 10 set |
| 4. | Manual, Hard copy in good quality paper properly bounded | 10 set |
| 5. | Copy of Type Test certificate along with manual | With offer |
| 6. | Basic application software for setting change, | 10 nos. |
| 7. | CD with software(licensed ) to download disturbance recorder, event log and evaluation of those records | 10 nos. |
| 8. | Graphical configuration tool for I/P, O/P and functional building block for protection | 10 nos. |
| 9. | Any other software required for integration with SCADA. | 10 nos. |

**N.B All the above tools/ Software should be compatible to WINDOWS XP/WINDOWS NT/WINDOWS 7 and latest WINDOWSOperating System.**

**SCHEDULE – I**



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## Undertaking from Relay manufacturer

We hereby confirm that the protective relay(s) type ____________________

i)
ii)
iii)

offered by us against your tender No.____________________ through M/s.______________________ are in our current range of production. We also confirm that these relays will not be phased out by us in the next 10 years from the date of supply. Necessary repairs/replacements if necessary during this period will be made available by us directly to WBSEDCL or through panel manufacturer.It is also confirm that as per WBSEDCL technical specification we will provide 5 years guarantee of all supplied relay directly to WBSEDCL or through panel manufacturer.

Name & Designation :

Company Seal :

We have offered our relay(s) type ______________________________

i)
ii)
iii)

to M/s. ________________________ against WBSEDCL's tender no.____________.

In this connection we hereby confirm that we would be extending all therequired technical support and back-up guarantee to WBSEDCL andM/s._______________ for the above mentioned relay(s).

Name & Designation :

Company Seal :

<u>Note: Copy of Schedule 1 shall also to be submitted during approval of drawing.</u>



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# ANNEXURE-F

## STANDARD TECHNICAL SPECIFCATION FOR CT/PT OPERATED DLMS COMPLIANT TRIVECTOR ENERGY METER FOR 11KV FEEDER METERING UPLOADED IN THE WBSEDCL WEBSITE TO BE FOLLOWED